Control Systems

Previous Year Questions • 1988–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Control Systems (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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UNIT

I

Basics of Control Systems

Topics

1. Introduction to Control Systems
2. Laplace Transform and Concept of a System
3. Transfer Function
4. Concept of Feedback, OLTF and Sensitivity
5. Poles, Zeros and Final Value Theorem
6. Dominant Pole, Impulse and Step Response

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Unit Snapshot*

Stream: EE	**Questions:** 12	**MCQ:** 11	**MSQ:** 0	**NAT:** 1
1M: 9	**2M:** 3	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2013)–3M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q1.20.1 MCQ 2020 1M

Consider a linear time-invariant system whose input $r(t)$ and output $y(t)$ are related by the following differential equation:

$$\frac{d^2y(t)}{dt^2} + 4y(t) = 6r(t)$$

The poles of this system are at

A. $+2j, -2j$ **B.** $+2, -2$
C. $+4, -4$ **D.** $+4j, -4j$

Q1.20.2 MCQ 2020 2M

A stable real linear time-invariant system with single pole at p, has a transfer function $H(s) = \frac{s^2 + 100}{s - p}$ with a dc gain of 5. The smallest positive frequency, in rad/s, at unity gain is closest to:

A. 8.84 **B.** 11.08 **C.** 78.13 **D.** 122.87

Q1.15.1 NAT 2015 1M

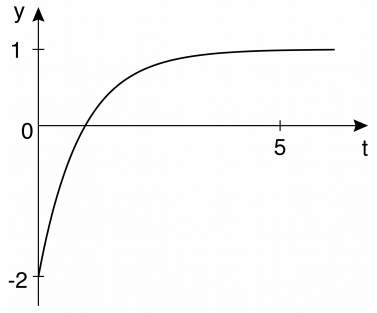
An open loop control system results in a response of $e^{-2t}(\sin 5t + \cos 5t)$ for a unit impulse input. The DC gain of the control system is

_____

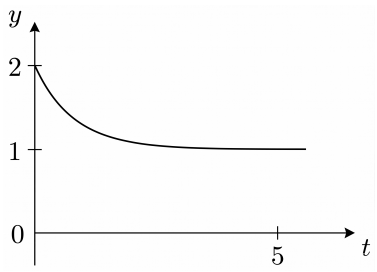
Q1.15.2 MCQ 2015 2M

The unit step response of a system with the transfer function $G(s) = \frac{1 - 2s}{1 + s}$ is given by which one of the following waveforms?

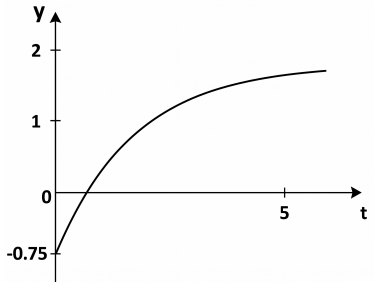
A.



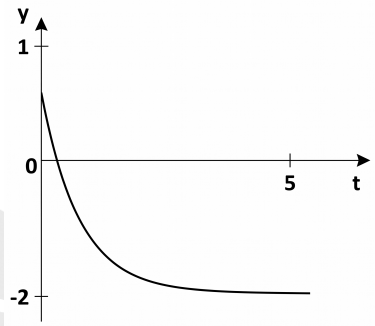
B.



C.



D.



Q1.13.1 MCQ 2013 1M

The transfer function $\frac{V_2(s)}{V_1(s)}$ of the circuit shown below is

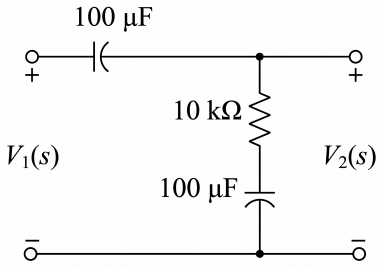


Fig. Q1.13.1

A. $\frac{0.5s + 1}{s + 1}$ **B.** $\frac{3s + 6}{s + 2}$ **C.** $\frac{s + 2}{s + 1}$ **D.** $\frac{s + 1}{s + 2}$

Q1.04.1 MCQ 2004 1M

For a tachometer if $\theta(t)$ is the rotor displacement is radians, $e(t)$ is the output voltage and K_t is the tachometer constant in V/rad/sec, then the transfer function, $\frac{E(s)}{\Theta(s)}$ will be

A. $K_t s^2$ **B.** $\frac{K_t}{s}$ **C.** $K_t s$ **D.** K_t

Q1.03.1 MCQ 2003 1M

A control system is defined by the following mathematical relationship

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 5x = 12\left(1 - e^{-2t}\right)$$

The response of the system as $t \to \infty$ is

A. $x = 6$ **B.** $x = 2$ **C.** $x = 2.4$ **D.** $x = -2$

Q1.03.2 MCQ 2003 2M

A control system with certain excitation is governed by the following mathematical equation

$$\frac{d^2x}{dt^2} + \frac{1}{2}\frac{dx}{dt} + \frac{1}{18}x = 10 + 5e^{-4t} + 2e^{-5t}$$

The natural time constants of the response of the system are

A. 2s and 5s **B.** 3s and 6s
C. 4s and 5s **D.** 1/3s and 1/6s

Q1.00.1 MCQ 2000 1M

Feedback control systems are

A. insensitive to both forward and feedback path parameter changes

B. less sensitive to feedback path parameter changes than to forward path parameter changes

C. less sensitive to forward-path parameter changes than to feedback path parameter changes

D. equally sensitive to forward and feedback path parameter changes

Q1.98.1 MCQ 1998 1M

The output of a linear time invariant control system is $c(t)$ for a certain input $r(t)$. If $r(t)$ is modified by passing it through a block whose transfer function is e^{-s} and then applied to the system, the modified output of the system would be

A. $\frac{c(t)}{1+e^{t}}$ **B.** $\frac{c(t)}{1+e^{-t}}$

C. $c(t-1)u(t-1)$ **D.** $c(t)u(t-1)$

Q1.96.1 MCQ 1996 1M

The unit-impulse response of a unit-feedback control system is given by

$$c(t) = -te^{-t} + 2e^{-t}, \quad (t \geq 0)$$

the open loop transfer function is equal to

A. $\frac{s+1}{(s+2)^2}$ **B.** $\frac{2s+1}{s^2}$

C. $\frac{s+1}{(s+1)^2}$ **D.** $\frac{s+1}{s^2}$

Q1.95.1 MCQ 1995 1M

The closed loop transfer function of a control system is given by

$$\frac{C(s)}{R(s)} = \frac{2(s-1)}{(s+2)(s+1)}$$

for a unit step input the output is

A. $-3e^{-2t} + 4e^{-t} - 1$ **B.** $-3e^{-2t} - 4e^{-t} + 1$
C. zero **D.** infinity

Answer Key

Q1.20.1 **A**	Q1.20.2 **A**	Q1.15.1 **0.23 to 0.25**
Q1.15.2 **A**	Q1.13.1 **D**	Q1.04.1 **C**
Q1.03.1 **C**	Q1.03.2 **B**	Q1.00.1 **C**
Q1.98.1 **C**	Q1.96.1 **B**	Q1.95.1 **A**

UNIT

II

Block Diagram and Signal Flow Graph

Topics

1. Block Diagram Reduction Techniques
2. Signal Flow Graphs and Mason's Gain Formula
3. Limitations of Mason's Gain Formula

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Unit Snapshot*

Stream: EE	**Questions:** 17	**MCQ:** 16	**MSQ:** 0	**NAT:** 1
1M: 10	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 0M(2014)–2M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q2.24.1 MCQ 2024 1M

For the block-diagram shown in the figure, the transfer][2pt] function $\frac{C(s)}{R(s)}$ is

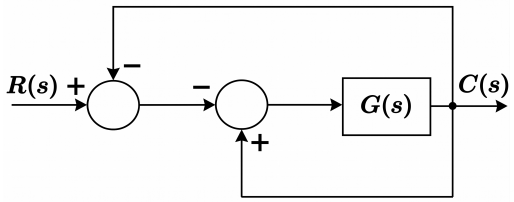


Fig. Q2.24.1

A. $\frac{G(s)}{1+2G(s)}$ **B.** $-\frac{G(s)}{1+2G(s)}$

C. $\frac{G(s)}{1-2G(s)}$ **D.** $-\frac{G(s)}{1-2G(s)}$

Q2.23.1 MCQ 2023 1M

For the block diagram shown in the figure, the transfer function $\frac{Y(s)}{R(s)}$ is

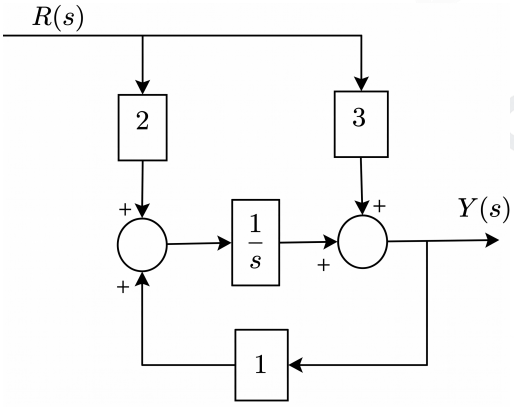


Fig. Q2.23.1

A. $\frac{2s+3}{s+1}$ **B.** $\frac{3s+2}{s-1}$ **C.** $\frac{s+1}{3s+2}$ **D.** $\frac{3s+2}{s+1}$

Q2.21.1 MCQ 2021 1M

For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

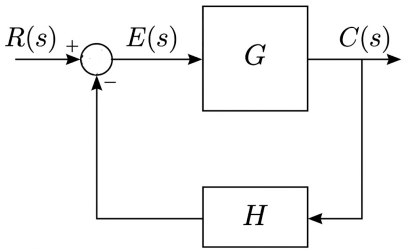


Fig. Q2.21.1

A. $\frac{G}{1+GH}$ **B.** $\frac{GH}{1+GH}$ **C.** $\frac{1}{1+GH}$ **D.** $\frac{1}{1+G}$

Q2.20.1 MCQ 2020 1M

Which of the options is an equivalent representation of the signal flow graph shown here?

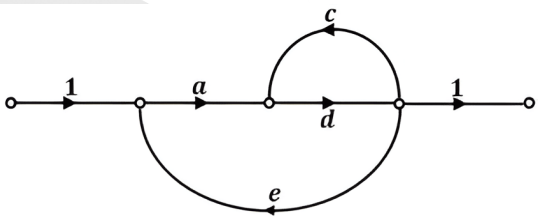
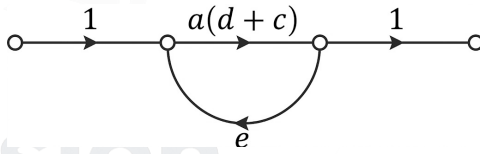
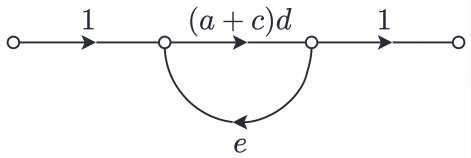
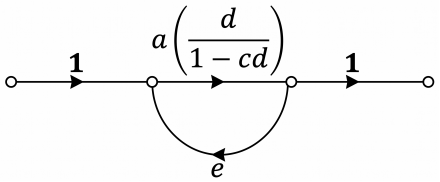
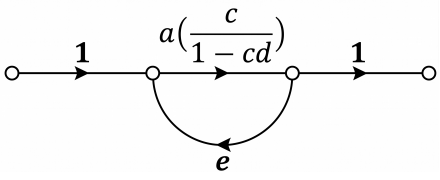


Fig. Q2.20.1

A.


B.


C.


D.


Q2.17.1 MCQ 2017 2M

In the system whose signal flow graph is shown in the figure, $U_1(s)$ and $U_2(s)$ are inputs. The transfer function $\frac{Y(s)}{U_1(s)}$ is

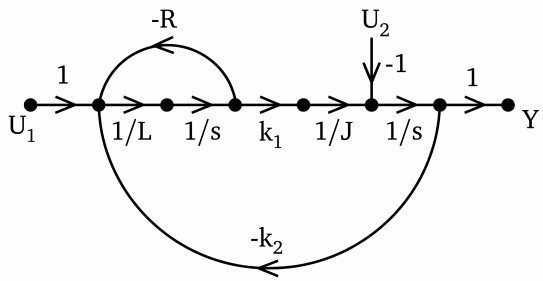


Fig. Q2.17.1

A. $\frac{k_1}{JLs^2 + JRs + k_1k_2}$

B. $\frac{k_1}{JLs^2 - JRs - k_1k_2}$

C. $\frac{k_1 - U_2(R + sL)}{JLs^2 + (JR - U_2L)s + k_1k_2 - U_2R}$

D. $\frac{k_1 - U_2(sL - R)}{JLs^2 - (JR + U_2L)s - k_1k_2 + U_2R}$

Q2.15.1 MCQ 2015 1M

For the signal-flow graph shown in the figure, which one of the following expressions is equal to the transfer function $\left.\frac{Y(s)}{X_2(s)}\right|_{X_1(s)=0}$?

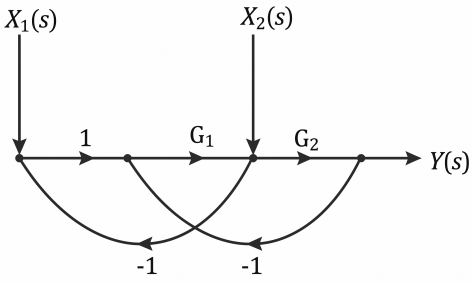


Fig. Q2.15.1

A. $\frac{G_1}{1 + G_2(1 + G_1)}$

B. $\frac{G_2}{1 + G_1(1 + G_2)}$

C. $\frac{G_1}{1 + G_1G_2}$

D. $\frac{G_2}{1 + G_1G_2}$

Q2.15.2 MCQ 2015 2M

Find the transfer function $\frac{Y(s)}{X(s)}$ of the system given below.

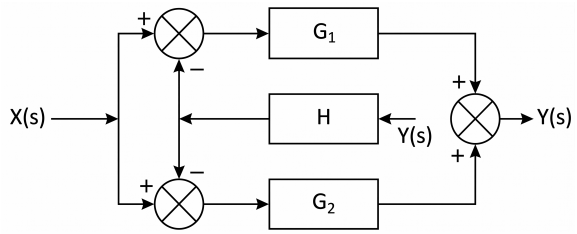


Fig. Q2.15.2

A. $\frac{G_1}{1 - HG_1} + \frac{G_2}{1 - HG_2}$

B. $\frac{G_1}{1 + HG_1} + \frac{G_2}{1 + HG_2}$

C. $\frac{G_1 + G_2}{1 + H(G_1 + G_2)}$

D. $\frac{G_1 + G_2}{1 - H(G_1 + G_2)}$

Q2.14.1 MCQ 2014 1M

The signal flow graph of a system is shown below. $U(s)$ is the input and $C(s)$ is the output.

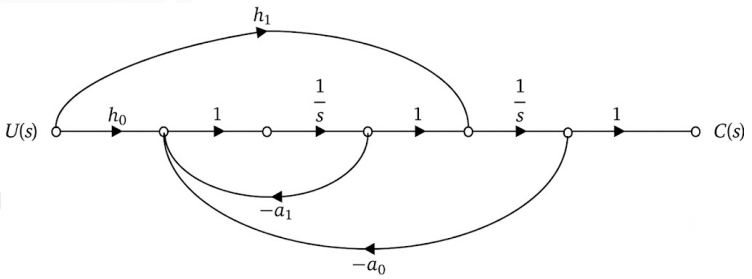


Fig. Q2.14.1

Assuming, $h_1 = b_1$ and $h_0 = b_0 - b_1a_1$, the input-output transfer function, $G(s) = \frac{C(s)}{U(s)}$ of the system is given by

A. $G(s) = \frac{b_0s + b_1}{s^2 + a_0s + a_1}$

B. $G(s) = \frac{a_1s + a_0}{s^2 + b_1s + b_0}$

C. $G(s) = \frac{b_1s + b_0}{s^2 + a_1s + a_0}$

D. $G(s) = \frac{a_0s + a_1}{s^2 + b_0s + b_1}$

Q2.13.1 MCQ 2013 2M

The signal flow graph for a system is given below. The transfer function $\frac{Y(s)}{U(s)}$ for this system is

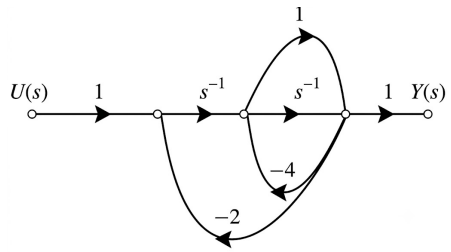


Fig. Q2.13.1

A. $\dfrac{s+1}{5s^2+6s+2}$ **B.** $\dfrac{s+1}{s^2+6s+2}$

C. $\dfrac{s+1}{s^2+4s+2}$ **D.** $\dfrac{1}{5s^2+6s+2}$

Q2.10.1 MCQ 2010 1M

As shown in the figure, a negative feedback system has an amplifier of gain 100 with ±10% tolerance in the forward path, and an attenuator of value 9/100 in the feedback path. The overall system gain is approximately:

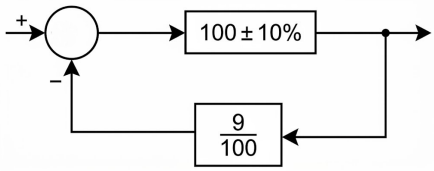


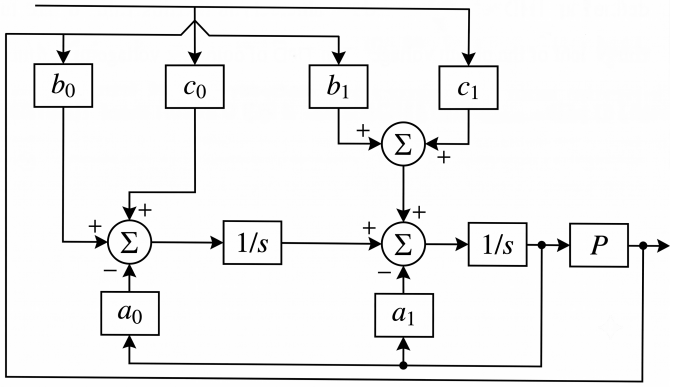
Fig. Q2.10.1

A. 10 ± 1% **B.** 10 ± 2%

C. 10 ± 5% **D.** 10 ± 10%

Q2.07.1 MCQ 2007 2M

The system shown in figure below



can be reduced to the form

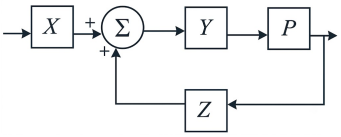


Fig. Q2.07.1

A. $X = c_0 s + c_1, Y = \dfrac{1}{(s^2 + a_0 s + a_1)}, Z = b_0 s + b_1$

B. $X = 1, Y = \dfrac{(c_0 s + c_1)}{(s^2 + a_0 s + a_1)}, Z = b_0 s + b_1$

C. $X = c_1 s + c_0, Y = \dfrac{(b_1 s + b_0)}{(s^2 + a_1 s + a_0)}, Z = 1$

D. $X = c_1 s + c_0, Y = \dfrac{1}{(s^2 + a_1 s + a_0)}, Z = b_1 s + b_0$

Q2.04.1 MCQ 2004 2M

For the block diagram shown in figure, the transfer function $\dfrac{C(s)}{R(s)}$ is equal to

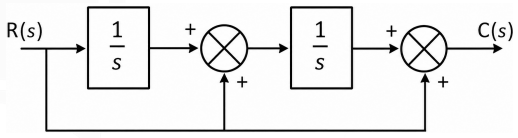


Fig. Q2.04.1

A. $\dfrac{s^2+1}{s^2}$ **B.** $\dfrac{s^2+s+1}{s^2}$

C. $\dfrac{1}{s^2+s+1}$ **D.** $\dfrac{s^2+s+1}{s}$

Q2.03.1 MCQ 2003 2M

The block diagram of a control system is shown in Fig. The transfer function $G(s) = \dfrac{Y(s)}{U(s)}$ of the system is

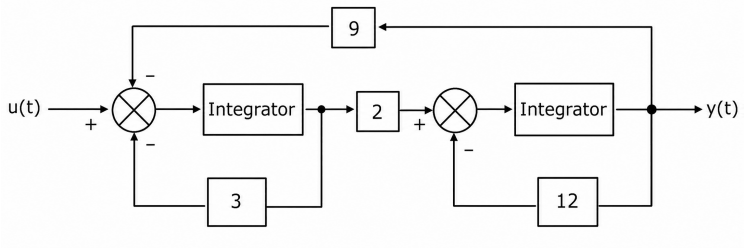


Fig. Q2.03.1

A. $\dfrac{1}{18\left(1+\frac{s}{12}\right)\left(1+\frac{s}{3}\right)}$ **B.** $\dfrac{1}{27\left(1+\frac{s}{6}\right)\left(1+\frac{s}{9}\right)}$

C. $\dfrac{1}{27\left(1+\frac{s}{12}\right)\left(1+\frac{s}{9}\right)}$ **D.** $\dfrac{1}{27\left(1+\frac{s}{9}\right)\left(1+\frac{s}{3}\right)}$

Q2.98.1 MCQ 1998 1M

For block diagram shown in Fig. $\dfrac{C(s)}{R(s)}$ is given by

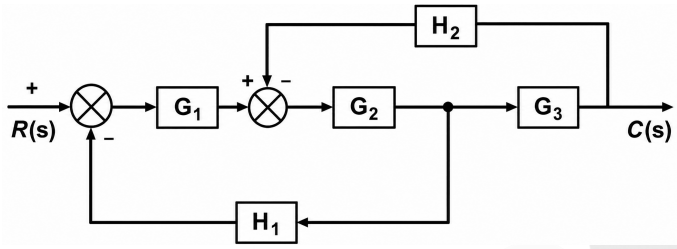


Fig. Q2.98.1

A. $\dfrac{G}{1-GH}$ **B.** $\dfrac{2G}{1-GH}$

C. $\dfrac{GH}{1-GH}$ **D.** $\dfrac{GH}{1-H}$

Q2.91.1 NAT 1991 2M

The signal flow graph of figure shown below, has _____ forward paths and _____ feedback loops.

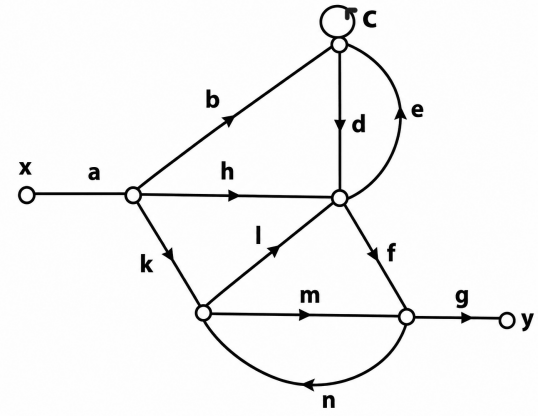


Fig. Q2.91.1

Q2.93.1 MCQ 1993 1M

Signal flow graph is used to obtain the

A. stability of a system

B. transfer function of a system

C. controllability of a system

D. observability of a system

Q2.92.1 MCQ 1992 1M

The overall transfer function of the system shown in figure is

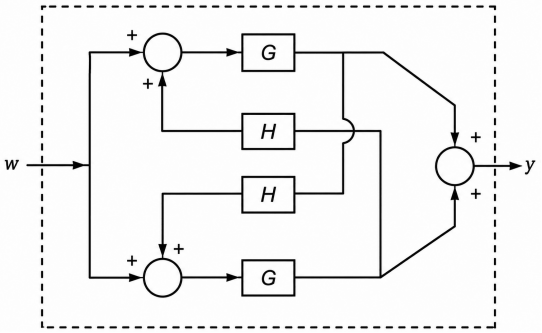


Fig. Q2.92.1

Answer Key

Q2.24.1 D	Q2.23.1 B	Q2.21.1 C
Q2.20.1 C	Q2.17.1 A	Q2.15.1 B
Q2.15.2 C	Q2.14.1 C	Q2.13.1 A
Q2.10.1 A	Q2.07.1 D	Q2.04.1 B
Q2.03.1 B	Q2.98.1 A	Q2.93.1 B
Q2.92.1 B	Q2.91.1 4,4	

PrepFusion

UNIT

III

Time Response Analysis

Topics

1. Impulse, Step, Ramp and Parabolic Response
2. Concept of Damping in 2nd Order Systems
3. Complex Frequency and Damping in RLC Circuits
4. Step Response of 2nd Order Systems
5. Time Domain Performance Parameters
6. Steady State Error — Unity and Non-Unity Feedback
7. Error Constants and System Types

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 36	**MCQ:** 28	**MSQ:** 0	**NAT:** 8
1M: 13	**2M:** 20	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 0M(2014)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 1M

The Laplace transform of the step response of a system is given by

$$Y(s) = \frac{100}{s(s+100)}$$

The rise time is defined as the time taken for the response to go from 0.1 to 0.9 of its final value. The settling time is defined as the time taken for the response to reach 0.98 of its final value.

For this system, the rise time (T_r), settling time (T_s), and time constant (T_c), all expressed in seconds, are

A. $T_r = 0.022, T_s = 0.04, T_c = 0.01$

B. $T_r = 0.22, T_s = 0.404, T_c = 0.01$

C. $T_r = 2.2, T_s = 4.04, T_c = 1.01$

D. $T_r = 22, T_s = 40.4, T_c = 10.1$

Q3.25.1 MCQ 2025 1M

Selected data points of the step response of a stable first-order linear time-invariant (LTI) system are given below. The closest value of the time-constant, in sec, of the system is

Time (sec)	0.6	1.6	2.6	10	∞
Output	0.78	1.65	2.18	2.98	3

A. 1 **B.** 2 **C.** 3 **D.** 4

Q3.24.1 MCQ 2024 1M

Consider the standard second-order system of the form $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ with the poles p and p^* having negative real parts. The pole locations are also shown in the figure.

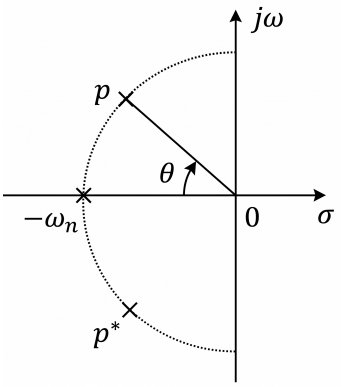


Fig. Q3.24.1

Now consider two such second-order systems as defined below:
System 1: $\omega_n = 3$ rad/sec and $\theta = 60°$
System 2: $\omega_n = 1$ rad/sec and $\theta = 70°$

Which one of the following statements is correct?

A. Settling time of System 1 is more than that of System 2.

B. Settling time of System 2 is more than that of System 1.

C. Settling times of both the systems are the same.

D. Settling time cannot be computed from the given information.

Q3.24.2 MCQ 2024 1M

Consider the cascaded system as shown in the figure. Neglecting the faster component of the transient response, which one of the following options is a first-order pole-only approximation such that the steady-state values of the unit step responses of the original and the approximated systems are same?

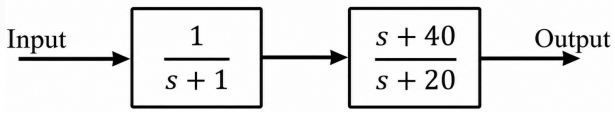


Fig. Q3.24.2

A. $\frac{1}{s+1}$ **B.** $\frac{2}{s+1}$ **C.** $\frac{1}{s+20}$ **D.** $\frac{2}{s+20}$

Q3.22.1 MCQ 2022 2M

The damping ratio and undamped natural frequency of a closed loop system as shown in the figure, are denoted as ζ and ω_n, respectively. The values of ζ and ω_n are

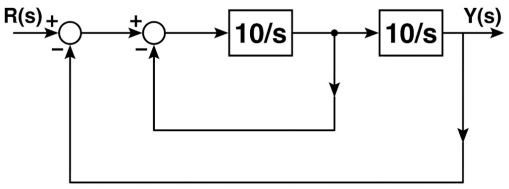


Fig. Q3.22.1

A. $\zeta = 0.5$ and $\omega_n = 10$ rad/s

B. $\zeta = 0.1$ and $\omega_n = 10$ rad/s

C. $\zeta = 0.707$ and $\omega_n = 10$ rad/s

D. $\zeta = 0.707$ and $\omega_n = 100$ rad/s

Q3.20.1 MCQ 2020 2M

Consider a permanent magnet dc (PMDC) motor which is initially at rest. At $t = 0$, a dc voltage of 5 V is applied to the motor. Its speed monotonically increases from 0 rad/s to 6.32 rad/s in 0.5 s and finally settles to 10 rad/s. Assuming that the armature inductance of the motor is negligible, the transfer function for the motor is

A. $\frac{10}{0.5s+1}$ **B.** $\frac{2}{0.5s+1}$ **C.** $\frac{2}{s+0.5}$ **D.** $\frac{10}{s+0.5}$

Q3.18.1 MCQ 2018 1M

Match the transfer functions of the second-order systems with the nature of the systems given below.

Transfer functions	*Nature of system*
P: $\frac{15}{s^2+5s+15}$	I: Overdamped
Q: $\frac{25}{s^2+10s+25}$	II: Critically damped
R: $\frac{35}{s^2+18s+35}$	III: Underdamped

A. P-I, Q-II, R-III **B.** P-II, Q-I, R-III
C. P-III, Q-II, R-I **D.** P-III, Q-I, R-II

Q3.18.2 NAT 2018 1M

Consider a unity feedback system with forward transfer function given by

$$G(s) = \frac{1}{(s+1)(s+2)}$$

The steady-state error in the output of the system for a unit-step input is (up to 2 decimal places)

Q3.18.3 NAT 2018 2M

The unit step response $y(t)$ of a unity feedback system with open loop transfer function $G(s)H(s) = \frac{K}{(s+1)^2(s+2)}$ is shown in the figure. The value of K is (up to 2 decimal places).

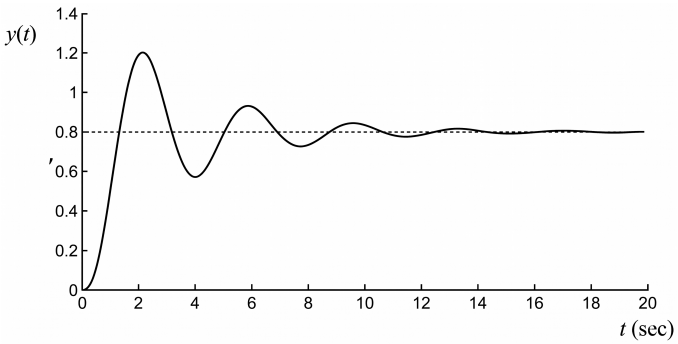


Fig. Q3.18.3

Q3.17.1 NAT 2017 2M

For a system having transfer function $G(s) = \frac{-s+1}{s+1}$, a unit step input is applied at time $t = 0$. The value of the response of the system at $t = 1.5$ sec (rounded off to three decimal places) is

Q3.17.2 MCQ 2017 1M

When a unit ramp input is applied to the unity feedback system having closed loop transfer function $\frac{C(s)}{R(s)} = \frac{Ks+b}{s^2+as+b}$, $(a > 0, b > 0, K > 0)$, the steady state error will be

A. 0 **B.** $\frac{a}{b}$ **C.** $\frac{a+K}{b}$ **D.** $\frac{a-K}{b}$

Q3.17.3 MCQ 2017 2M

Which of the following systems has maximum peak overshoot due to a unit step input?

A. $\frac{100}{s^2+10s+100}$ **B.** $\frac{100}{s^2+15s+100}$

C. $\frac{100}{s^2+5s+100}$ **D.** $\frac{100}{s^2+20s+100}$

Q3.16.1 MCQ 2016 2M

A second-order real system has the following properties:
a) the damping ratio $\zeta = 0.5$ and undamped natural frequency $\omega_n = 10$ rad/s,
b) the steady state value of the output, to a unit step input, is 1.02.
The transfer function of the system is

A. $\dfrac{1.02}{s^2+5s+100}$ **B.** $\dfrac{102}{s^2+10s+100}$

C. $\dfrac{100}{s^2+10s+100}$ **D.** $\dfrac{102}{s^2+5s+100}$

Q3.14.1 NAT 2014 1M

The closed-loop transfer function of a system is $T(s) = \dfrac{4}{(s^2+0.4s+4)}$. The steady state error due to unit step input is

---

Q3.13.1 MCQ 2013 2M

The open-loop transfer function of a dc motor is given as $\dfrac{\omega(s)}{V_a(s)} = \dfrac{10}{1+10s}$. When connected in feedback as shown below, the approximate value of K_a that will reduce the time constant of the closed loop system by one hundred times as compared to that of the open-loop system is

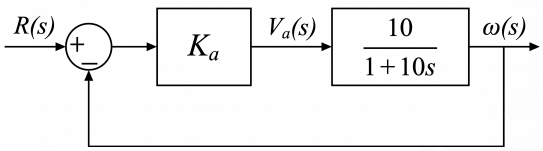


Fig. Q3.13.1

A. 1 **B.** 5 **C.** 10 **D.** 100

Q3.11.1 MCQ 2011 1M

The steady-state error of a unity feedback linear system for a unit step input is 0.1. The steady-state error of the same system, for a pulse input $r(t)$ having a magnitude of 10 and a duration of one second as shown in the figure is

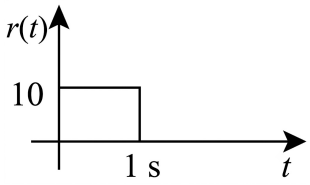


Fig. Q3.11.1

A. 0 **B.** 0.1 **C.** 1 **D.** 10

Q3.11.2 MCQ 2011 2M

A two-loop position control system is shown below.

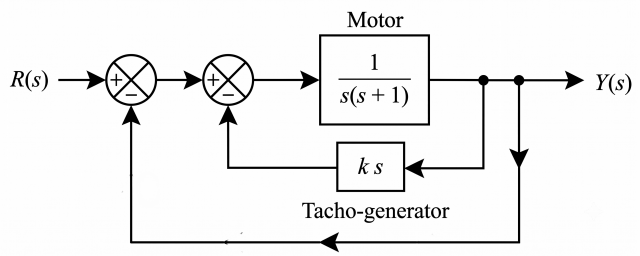


Fig. Q3.11.2

The gain k of the Tacho-generator influences mainly the

A. peak overshoot

B. natural frequency of oscillation

C. phase shift of the closed loop transfer function at very low frequencies ($\omega \to 0$)

D. phase shift of the closed loop transfer function at very high frequencies ($\omega \to \infty$)

Q3.10.1 MCQ 2010 1M

For the system $\dfrac{2}{(s+1)}$, the approximate time taken for a step response to reach 98% of its final value is

A. 1 s **B.** 2 s **C.** 4 s **D.** 8 s

Q3.09.1 MCQ 2009 2M

The unit-step response of a unity feedback system with open loop transfer function $G(s) = \dfrac{K}{(s+1)(s+2)}$ is shown in the figure. The value of K is

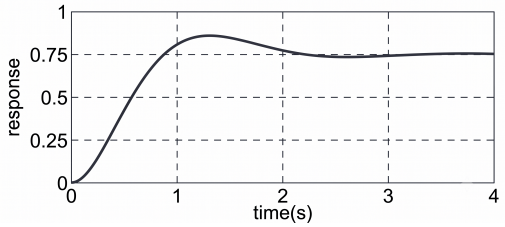


Fig. Q3.09.1

A. 0.5 **B.** 2 **C.** 4 **D.** 6

Q3.08.1 MCQ 2008 2M

The transfer function of a linear time invariant system is given as

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

The steady state value of the output of this system for a unit impulse input applied at time instant $t = 1$ will be

A. 0 **B.** 0.5 **C.** 1 **D.** 2

Q3.07.1 MCQ 2007 2M

Consider the feedback control system shown below which is subjected to a unit step input. The system is stable and has the following parameters $k_p = 4$, $k_i = 10$, $\omega = 500$ and $\xi = 0.7$.

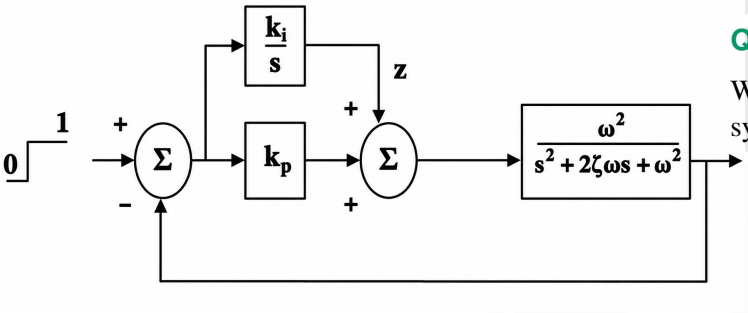


Fig. Q3.07.1

The steady state value of z is

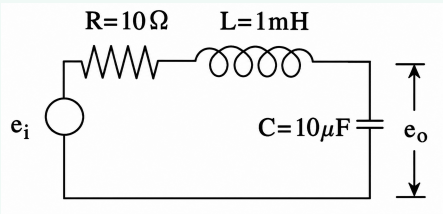
A. 1 **B.** 0.25 **C.** 0.1 **D.** 0

Common Data Questions

The next 2 questions share the data below.

Common data for Q3.07.2 and Q3.07.3

Consider the R-L-C circuit shown in figure.



Q3.07.2 MCQ 2007 2M

For a step-input e_i, the overshoot in the output e_o will be

A. 0, since the system is not under-damping **B.** 5%
C. 16% **D.** 48%

Q3.07.3 MCQ 2007 2M

If the above step response is to be observed on a non-storage CRO, then it would be best to have the e_i as a

A. step function

B. square wave of frequency 50 Hz

C. square wave of frequency 300 Hz

D. square wave of frequency 2.0 kHz

Q3.05.1 MCQ 2005 2M

When subjected to a unit step input, the closed loop control system shown in figure will have a steady state error of

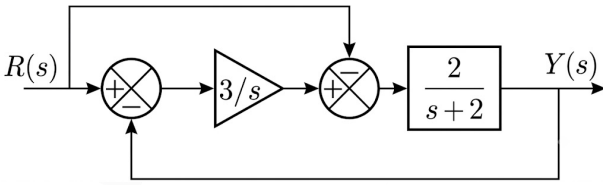


Fig. Q3.05.1

A. −1.0 **B.** −0.5 **C.** 0 **D.** 0.5

Q3.04.1 MCQ 2004 2M

The unit impulse response of a second order under-dumped system starting from rest is given by

$$c(t) = 12.5e^{-6t} \sin 8t, t \geq 0$$

The steady-state value of the unit step response of the system is equal to

A. 0 **B.** 0.25 **C.** 0.5 **D.** 1.0

Q3.04.2 MCQ 2004 2M

The block diagram of a closed loop control system is given by figure. The values of K and P such that the system has a damping ratio of 0.7 and an undamped natural frequency ω_n of 5 rad/sec, are respectively equal to

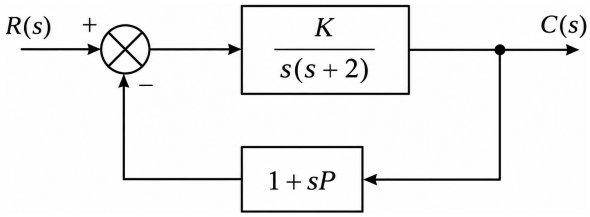


Fig. Q3.04.2

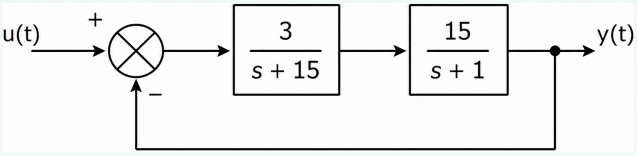
A. 20 and 0.3 **B.** 20 and 0.2
C. 25 and 0.3 **D.** 25 and 0.2

Common Data Questions

The next 2 questions share the data below.

Common data for Q3.03.1 and Q3.03.2

The block diagram shown in Fig. gives a unity feedback closed loop control system.



Q3.03.1 MCQ 2003 2M

The steady state error in the response of the above system to unit step input is

A. 25% **B.** 0.75% **C.** 6% **D.** 33%

Q3.03.2 MCQ 2003 2M

The roots of the closed loop characteristic equation of the system shown in above Fig. are

A. −1 and −15 **B.** 6 and 10
C. −4 and −15 **D.** −6 and −10

Q3.00.1 MCQ 2000 1M

A unity feedback system has open-loop transfer function $G(s)$. The steady-state error is zero for

A. step input and type-1 $G(s)$

B. ramp input and type-1 $G(s)$

C. step input and type-0 $G(s)$

D. ramp input and type-0 $G(s)$

Q3.00.2 MCQ 2000 2M

A unity feedback system has open-loop transfer function $G(s) = \dfrac{25}{[s(s+6)]}$. The peak overshoot in the step-input response of the system is approximately equal to:

A. 5% **B.** 10% **C.** 15% **D.** 20%

Q3.97.1 NAT 1997 5M

A first order system is initially at rest and excited by a step input at time $t = 0$. Its output becomes 1.1 V is in 4 seconds and eventually reaches a steady state value of 2 V. Determine its time constant_____

Q3.96.1 MCQ 1996 1M

For a feedback control system of type 2, the steady state error for a ramp input is

A. infinite **B.** constant
C. zero **D.** indeterminate

Q3.95.1 NAT 1995 5M

Match the following root locations of the characteristic Equations of second order systems with their corresponding Unit step responses of second order systems.

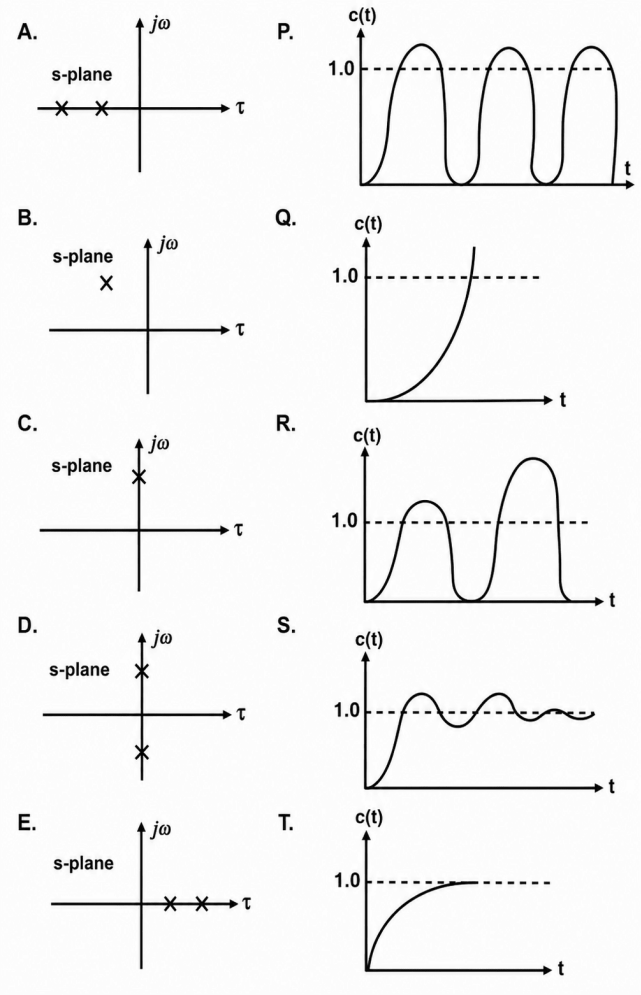


Fig. Q3.95.1

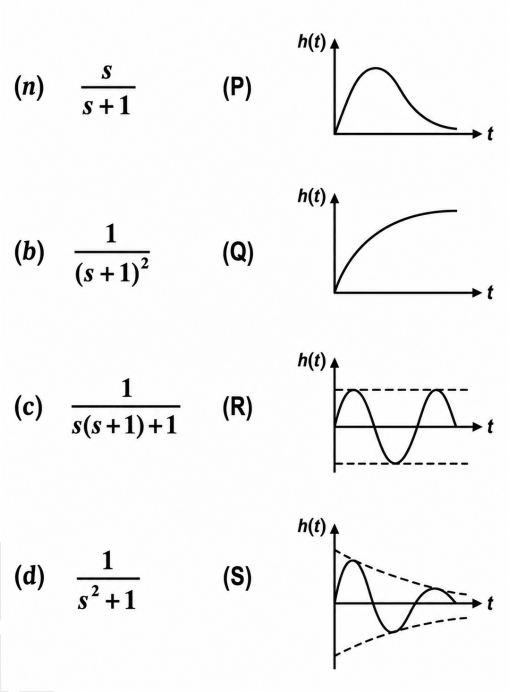


Fig. Q3.92.1

Q3.92.2 MCQ | 1992 | 1M

For what values of 'a' does the system shown in figure have a zero steady state error [i.e. $\lim_{t\to\infty} E(t)$] for a step input?

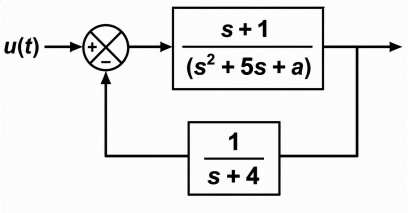


Fig. Q3.92.2

A. $a = 0$
B. $a = 2$
C. $a \geq 4$
D. for no value of 'a'

Q3.92.1 NAT | 1992 | 4M

Match the following transfer functions and impulse responses

Q3.91.1 NAT | 1991 | 2M

A first order system and its response to a unit step input are shown in the figure below. The system parameters are $a =$ _____, $K =$ _____

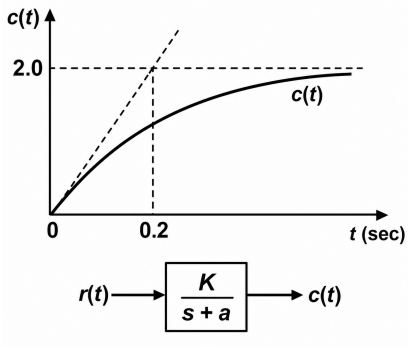


Fig. Q3.91.1

Answer Key

Note: **MTA** Marks To All awarded — see detailed solutions.

Q3.26.1 A	Q3.25.1 B	Q3.24.1 B
Q3.24.2 B	Q3.22.1 A	Q3.20.1 B
Q3.18.1 C	Q3.18.2 0.65 to 0.69	Q3.18.3 8
Q3.17.1 0.550 to 0.556	Q3.17.2 D	Q3.17.3 C
Q3.16.1 B	Q3.14.1 0	Q3.13.1 C
Q3.11.1 A	Q3.11.2 A	Q3.10.1 C
Q3.09.1 D	Q3.08.1 A	Q3.07.1 A
Q3.07.2 C	Q3.07.3 C	Q3.05.1 C
Q3.04.1 D	Q3.04.2 D	Q3.03.1 A
Q3.03.2 D	Q3.00.1 A	Q3.00.2 B
Q3.97.1 5	Q3.96.1 C	Q3.95.1 A-T, B-S, C-P, D-R, E-Q
Q3.92.1 MTA	Q3.92.2 A	Q3.91.1 5,10

UNIT

IV

Routh's Stability Criterion

Topics

1. Hurwitz Polynomial and Routh Array
2. Special Cases in Routh Stability
3. Mapping Theorem in Routh-Hurwitz

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 34	**MCQ:** 22	**MSQ:** 0	**NAT:** 12
1M: 9	**2M:** 21	**Descriptive:** 4	**Avg Weightage** ~2M	**Range:** 1M(2022)–4M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q4.25.1 MCQ 2025 2M

Let $G(s) = \dfrac{1}{(s+1)(s+2)}$. Then the closed-loop system shown in the figure below, is

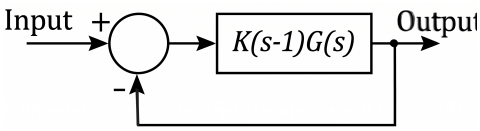


Fig. Q4.25.1

A. stable for all $K > 2$.

B. unstable for all $K > 2$.

C. unstable for all $K > 1$.

D. stable for all $K > 1$.

Q4.22.1 MCQ 2022 1M

The open loop transfer function of a unity gain negative feedback system is given by $G(s) = \dfrac{k}{s^2 + 4s - 5}$. The range of k for which the system is stable, is

A. $k > 3$ **B.** $k < 3$ **C.** $k > 5$ **D.** $k < 5$

Q4.20.1 MCQ 2020 2M

Which of the following options is correct for the system shown below?

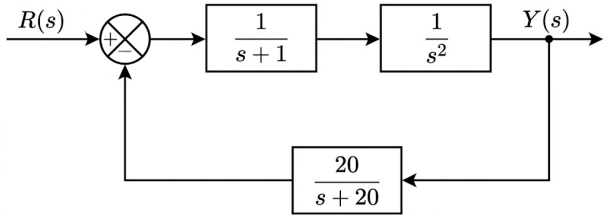


Fig. Q4.20.1

A. 4th order and stable **B.** 3rd order and stable
C. 4th order and unstable **D.** 3rd order and unstable

Q4.20.2 MCQ 2020 2M

Consider a negative unity feedback system with the forward path transfer function $\dfrac{s^2 + s + 1}{s^3 + 2s^2 + 2s + K}$, where K is a positive real number. The value of K for which the system will have some of its poles on the imaginary axis is

A. 9 **B.** 8 **C.** 7 **D.** 6

Q4.19.1 MCQ 2019 1M

The characteristic equation of a linear time-invariant (LTI) system is given by $\Delta(s) = s^4 + 3s^3 + 3s^2 + s + k = 0$. The system is BIBO stable if

A. $0 < k < \dfrac{12}{9}$ **B.** $k > 3$

C. $0 < k < \dfrac{8}{9}$ **D.** $k > 6$

Q4.18.1 MCQ 2018 2M

The number of roots of the polynomial, $s^7 + s^6 + 7s^5 + 14s^4 + 31s^3 + 73s^2 + 25s + 200$, in the open left half of the complex plane is

A. 3 **B.** 4 **C.** 5 **D.** 6

Q4.17.1 MCQ 2017 1M

A closed loop system has the characteristic equation given by $s^3 + Ks^2 + (K+2)s + 3 = 0$. For this system to be stable, which one of the following conditions should be satisfied?

A. $0 < K < 0.5$ **B.** $0.5 < K < 1$
C. $0 < K < 1$ **D.** $K > 1$

Q4.17.2 MCQ 2017 2M

The range of K for which all the roots of the equation $s^3 + 3s^2 + 2s + K = 0$ are in the left half of the complex s-plane is

A. $0 < K < 6$ **B.** $0 < K < 16$
C. $6 < K < 36$ **D.** $6 < K < 16$

Q4.16.1 NAT 2016 2M

Given the following polynomial equation

$$s^3 + 5.5s^2 + 8.5s + 3 = 0,$$

the number of roots of the polynomial, which have real parts strictly less than -1, is

---

Q4.16.2 MCQ 2016 2M

The open loop transfer function of a unity feedback control system is given by

$$G(s) = \frac{K(s+1)}{s(1+Ts)(1+2s)}, \quad K > 0, T > 0.$$

The closed loop system will be stable if,

A. $0 < T < \dfrac{4(K+1)}{K-1}$ **B.** $0 < K < \dfrac{4(T+2)}{T-2}$

C. $0 < K < \dfrac{T+2}{T-2}$ **D.** $0 < T < \dfrac{8(K+1)}{K-1}$

Q4.14.1 MCQ 2014 1M

In the formation of Routh-Hurwitz array for a polynomial, all the elements of a row have zero values. This premature termination of the array indicates the presence of

A. only one root at the origin **B.** imaginary roots
C. only positive real roots **D.** only negative real roots

Q4.14.2 NAT 2014 2M

For the given system, it is desired that the system be stable. The minimum value of α for this condition is

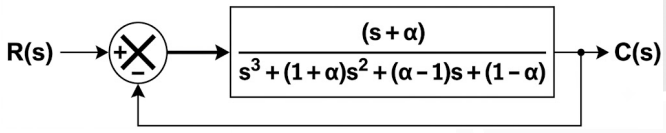


Fig. Q4.14.2

Q4.14.3 NAT 2014 2M

A system with the open loop transfer function

$$G(s) = \frac{K}{s(s+2)(s^2+2s+2)}$$

is connected in a negative feedback configuration with a feedback gain of unity. For the closed loop system to be marginally stable, the value of K is

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Q4.12.1 MCQ 2012 2M

The feedback system shown below oscillates at 2 rad/s when

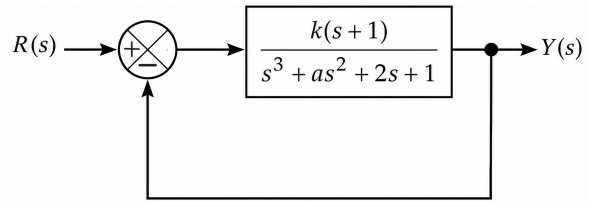


Fig. Q4.12.1

A. $K = 2$ and $a = 0.75$ **B.** $K = 3$ and $a = 0.75$
C. $K = 4$ and $a = 0.5$ **D.** $K = 2$ and $a = 0.5$

Q4.11.1 MCQ 2011 1M

An open loop system represented by the transfer function $G(s) = \dfrac{(s-1)}{(s+2)(s+3)}$ is

A. stable and of the minimum phase type

B. stable and of the non-minimum phase type

C. unstable and of the minimum phase type

D. unstable and of the non-minimum phase type

Q4.09.1 MCQ 2009 1M

The first two rows of Routh's tabulation of a third order equation are as follows.

s^3	2	2
s^2	4	4

This means there are

A. two roots at $s = \pm j$ and one root in right half s-plane

B. two roots at $s = \pm j2$ and one root in left half s-plane

C. two roots at $s = \pm j2$ and one root in right half s-plane

D. two roots at $s = \pm j$ and one root in left half s-plane

Q4.08.1 MCQ 2008 2M

Figure shows a feedback system where $K > 0$.

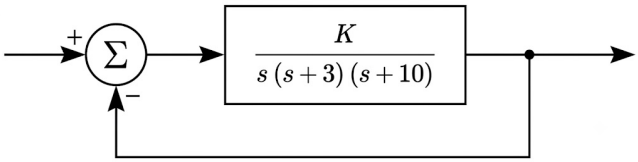


Fig. Q4.08.1

The range of K for which the system is stable will be given by

A. $0 < K < 30$ **B.** $0 < K < 39$
C. $0 < K < 390$ **D.** $K > 390$

Q4.07.1 MCQ 2007 1M

The system shown in the figure is

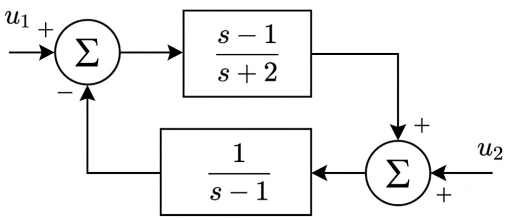


Fig. Q4.07.1

A. stable

B. unstable

C. conditionally stable

D. stable for input u_1, but unstable for input u_2

Q4.07.2 MCQ 2007 2M

If the loop gain K of a negative feedback system having a loop transfer function $\frac{K(s+3)}{(s+8)^2}$ is to be adjusted to induce a sustained oscillation then

A. The frequency of this oscillation is $4/\sqrt{3}$ rad/s

B. The frequency of this oscillation is 4 rad/s

C. The frequency of this oscillation is 4 or $4/\sqrt{3}$ rad/s

D. such a K does not exist

Q4.06.1 MCQ 2006 2M

The algebraic equation $F(s) = s^5 - 3s^4 + 5s^3 - 7s^2 + 4s + 20$ is given. $F(s) = 0$ has

A. a single complex root with the remaining roots being real

B. one positive real root and four complex roots, all with positive real parts

C. one negative real root, two imaginary roots, and two roots with positive real parts

D. one positive real root, two imaginary roots, and two roots with negative real parts

Q4.05.1 MCQ 2005 2M

A unity feedback system, having an open loop gain $G(s)H(s) = \frac{K(1-s)}{(1+s)}$, becomes stable when

A. $|K| > 1$ **B.** $K > 1$ **C.** $|K| < 1$ **D.** $K < -1$

Q4.04.1 MCQ 2004 2M

For the equation, $s^3 - 4s^2 + s + 6 = 0$, the number of roots in the left half of s-plane will be

A. zero **B.** one **C.** two **D.** three

Q4.03.1 MCQ 2003 2M

The loop gain GH of a closed system is given by the following expression $\frac{K}{s(s+2)(s+4)}$. The value of K for which the system just becomes unstable is

A. $K = 6$ **B.** $K = 8$ **C.** $K = 48$ **D.** $K = 96$

Q4.00.1 MCQ 2000 2M

The characteristic equation of a feedback control system is:

$$2s^4 + s^3 + 3s^2 + 5s + 10 = 0$$

The number of roots in the right half of s-plane are:

A. 0 **B.** 1 **C.** 2 **D.** 3

Q4.99.1 MCQ 1999 1M

None of the poles of a linear control system lie in the right half of s-plane. For a bounded input, the output of this system

A. is always bounded **B.** could be unbounded

C. always tends to zero **D.** None of the above

Q4.97.1 NAT 1997 2M

The system represented by the transfer function

$$G(s) = \frac{s^2 + 10s + 24}{s^4 + 6s^3 - 39s^2 + 19s + 84}$$

has how many pole (s) in the right-half s-plane.

_____

Q4.97.2 NAT 1997 5M

Determine whether the system given by the block diagram of Figure, is stable.

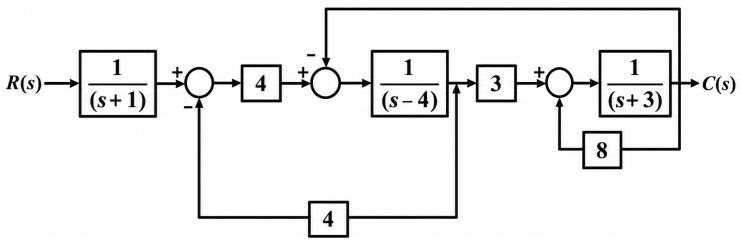


Fig. Q4.97.2

Q4.95.1 NAT 1995 2M

Closed loop stability implies that $[1 + G(s)H(s)]$ has _____ only in the left half of the s-plane.

Common Data Questions

The next 3 questions share the data below.

Common data for Q4.95.2, Q4.95.3, Q4.95.4 The characteristic equation for a certain feedback control system is given by $s^4 + 20s^3 + 15s^2 + 2s + K = 0$

Q4.95.2 NAT 1995 5M

Find the range of K for stability

_____

Q4.95.3 NAT 1995 5M

What is the frequency in rad/sec at which the system will oscillate?

_____

Q4.95.4 NAT 1995 5M

How many roots of the characteristic equation lie in the right half of the s-plane for $K = 5$?

_____

Q4.94.1 NAT 1994 1M

The number of positive real roots of the equation $s^3 - 2s + 2 = 0$ is

_____

Q4.94.2 NAT 1994 2M

The closed loop system, of Figure, is stable if the transfer function $T(s) = \dfrac{C(s)}{R(s)}$ is stable. (True/False)

_____

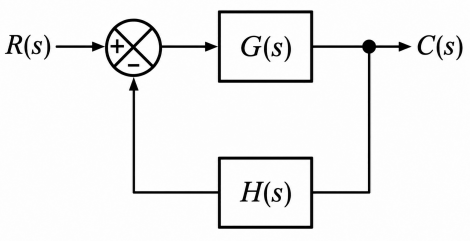


Fig. Q4.94.2

Q4.92.1 NAT 1992 2M

For what range of K is the following system (figure) asymptotically stable? Assume $K \geq 0$

_____

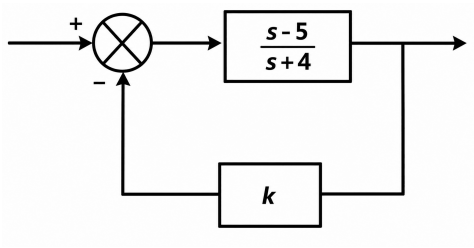


Fig. Q4.92.1

Answer Key

Q4.25.1 B	Q4.22.1 C	Q4.20.1 C
Q4.20.2 B	Q4.19.1 C	Q4.18.1 A
Q4.17.1 D	Q4.17.2 A	Q4.16.1 2
Q4.16.2 C	Q4.14.1 B	Q4.14.2 0.61 to 0.63
Q4.14.3 5	Q4.12.1 A	Q4.11.1 B
Q4.09.1 D	Q4.08.1 C	Q4.07.1 D
Q4.07.2 B	Q4.06.1 C	Q4.05.1 C
Q4.04.1 B	Q4.03.1 C	Q4.00.1 C
Q4.99.1 B	Q4.97.1 2	Q4.97.2 Stable
Q4.95.1 zeros	Q4.95.2 0<k<1.49	Q4.95.3 0.316 rad/sec
Q4.95.4 2	Q4.94.1 0	Q4.94.2 True
Q4.92.1 $0 \leq k < \frac{4}{5}$		

UNIT

V

Root Locus

Topics

1. Basics of Root Locus
2. Rules for Sketching Root Locus
3. Angle of Departure and Arrival
4. Complementary Root Locus

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 22	**MCQ:** 11	**MSQ:** 2	**NAT:** 9
1M: 5	**2M:** 9	**Descriptive:** 8	**Avg Weightage** ~2M	**Range:** 0M(2014)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q5.25.1 MSQ 2025 1M

The open-loop transfer function of the system shown in the figure, is $G(s) = \dfrac{Ks(s+2)}{(s+5)(s+7)}$ For $K \geq 0$, which of the following real axis point(s) is/are on the root locus?

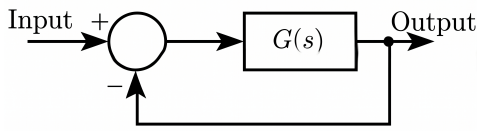


Fig. Q5.25.1

A. -1 **B.** -4 **C.** -6 **D.** -10

Q5.24.1 NAT 2024 2M

Consider the closed-loop system shown in the figure with

$$G(s) = \frac{K(s^2 - 2s + 2)}{(s^2 + 2s + 5)}.$$

The root locus for the closed-loop system is to be drawn for $0 \leq K < \infty$. The angle of departure (between $0°$ and $360°$) of the root locus branch drawn from the pole $(-1 + j2)$, in degrees, is (rounded off to the nearest integer)

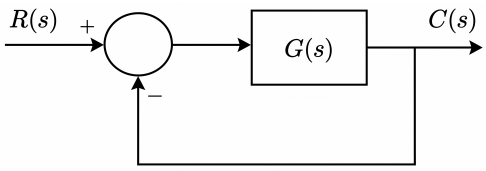


Fig. Q5.24.1

Q5.17.1 MCQ 2017 2M

The root locus of the feedback control system having the characteristic equation $s^2 + 6Ks + 2s + 5 = 0$ where $K > 0$, enters into the real axis at

A. $s = -1$ **B.** $s = -\sqrt{5}$ **C.** $s = -5$ **D.** $s = \sqrt{5}$

Q5.16.1 MCQ 2016 2M

The gain at the breakaway point of the root locus of a unity feedback system with open loop transfer function $G(s) = \dfrac{Ks}{(s-1)(s-4)}$ is

A. 1 **B.** 2 **C.** 5 **D.** 9

Q5.15.1 MCQ 2015 2M

The open loop poles of a third order unity feedback system are at $0, -1, -2$. Let the frequency corresponding to the point where the root locus of the system transits to unstable region be K. Now suppose we introduce a zero in the open loop transfer function at -3, while keeping all the earlier open loop poles intact. Which one of the following is TRUE about the point where the root locus of the modified system transits to unstable region?

A. It corresponds to a frequency greater than K

B. It corresponds to a frequency less than K

C. It corresponds to a frequency K

D. Root locus of modified system never transits to unstable region

Q5.15.2 MCQ 2015 2M

An open loop transfer function $G(s)$ of a system is

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

For a unity feedback system, the breakaway point of the root loci on the real axis occurs at,

A. -0.42 **B.** -1.58
C. -0.42 and -1.58 **D.** none of the above

Q5.14.1 MCQ 2014 1M

The root locus of a unity feedback system is shown in the figure

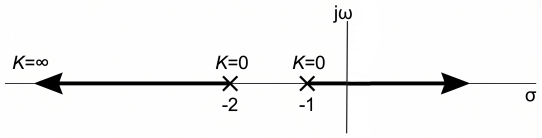


Fig. Q5.14.1

The closed loop transfer function of the system is

A. $\dfrac{C(s)}{R(s)} = \dfrac{K}{(s+1)(s+2)}$

B. $\dfrac{C(s)}{R(s)} = \dfrac{-K}{(s+1)(s+2)+K}$

C. $\dfrac{C(s)}{R(s)} = \dfrac{K}{(s+1)(s+2)-K}$

D. $\dfrac{C(s)}{R(s)} = \dfrac{K}{(s+1)(s+2)+K}$

Q5.11.1 MCQ 2011 2M

The open loop transfer function $G(s)$ of a unity feedback control system is given as,

$$G(s) = \frac{k\left(s + \frac{2}{3}\right)}{s^2(s+2)}.$$

From the root locus, it can be inferred that when k tends to positive infinity,

A. three roots with nearly equal real parts exist on the left half of the s-plane

B. one real root is found on the right half of the s-plane

C. the root loci cross the $j\omega$ axis for a finite value of k; $k \neq 0$

D. three real roots are found on the right half of the s-plane

Q5.10.1 MCQ 2010 2M

The characteristic equation of a closed-loop system is $s(s+1)(s+3) + k(s+2) = 0,\ k > 0$. Which of the following statements is true?

A. Its roots are always real

B. It cannot have a breakaway point in the range $-1 < \text{Re}[s] < 0$

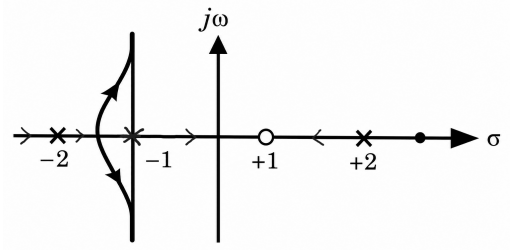
C. Two of its roots tend to infinity along the asymptotes $\text{Re}[s] = -1$

D. It may have complex roots in the right half plane.

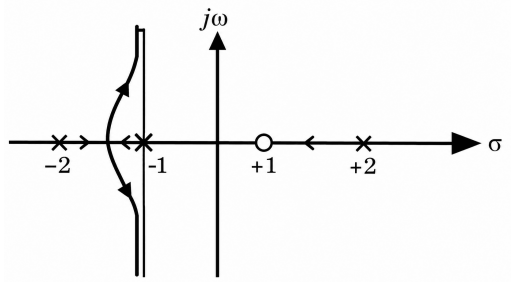
Q5.06.1 MCQ 2006 2M

A closed-loop system has the characteristic function $(s^2 - 4)(s+1) + K(s-1) = 0$. Its root locus plot against K is

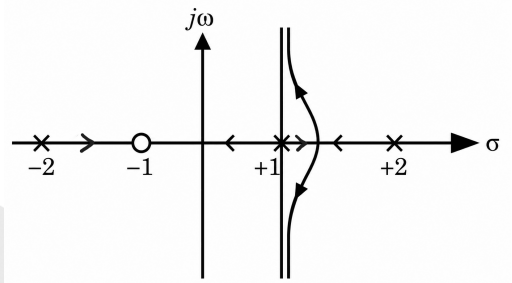
A.



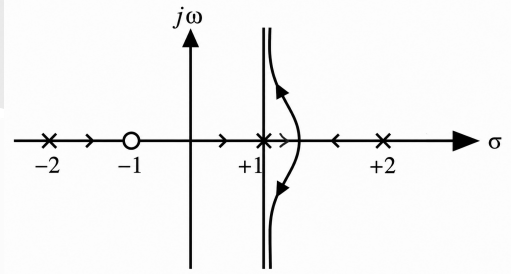
B.



C.



D.



Q5.05.1 MCQ 2005 1M

Below figure shows the root locus plot (location of poles not given) of a third order system whose open loop transfer function is:

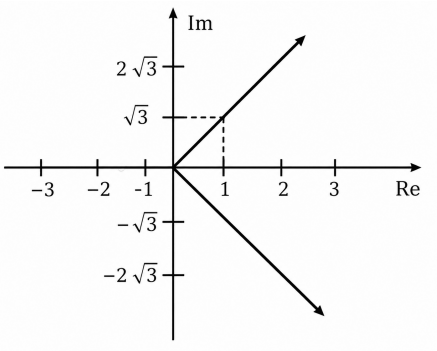
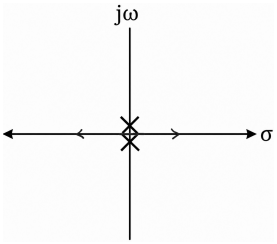


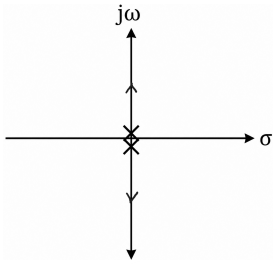
Fig. Q5.05.1

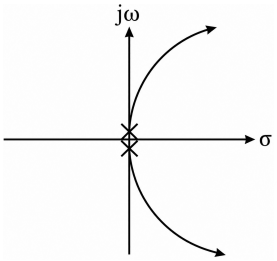
A. $\frac{K}{s^3}$ **B.** $\frac{K}{s^2(s+1)}$ **C.** $\frac{K}{s(s^2+1)}$ **D.** $\frac{K}{s(s^2-1)}$

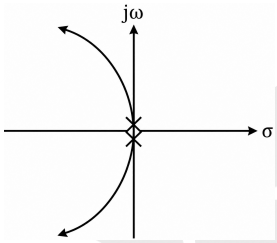
Q5.02.1 MCQ 2002 2M

A unity feedback system has an open loop transfer function, $G(s) = \frac{K}{s^2}$. The root locus plot is

A.


B.


C.


D.


Q5.02.2 NAT 2002 5M

The open loop transfer function of a unity feedback system is given by $G(s) = \frac{2(s+\alpha)}{s(s+2)(s+10)}$. Sketch the root locus as α varies from 0 to ∞. Find the angle and real axis intercept of the asymptotes, breakaway points and the imaginary axis crossing points, if any.

Q5.01.1 NAT 2001 5M

Given the characteristic equation

$$s^3 + 2s^2 + Ks + K = 0.$$

Sketch the root locus as K varies from zero to infinity. Find the angle and real axis intercept of the asymptotes, break-away/break-in points, and imaginary axis crossing points, if any.

Common Data Questions

The next 3 questions share the data below.

Common data for Q5.00.1, Q5.00.2, Q5.00.3

A unity feedback system has open-loop transfer function $G(s) = \frac{K(s+5)}{s(s+2)}; \quad K \geq 0$

Q5.00.1 NAT 2000 5M

Draw a rough sketch of the root locus plot; given that the complex roots of the characteristic equation move along a circle.

Q5.00.2 NAT 2000 5M

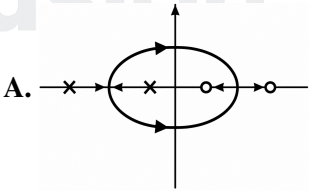
As K increases, does the system become less stable? Justify your answer.

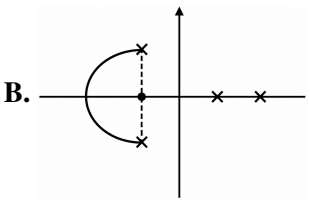
Q5.00.3 NAT 2000 5M

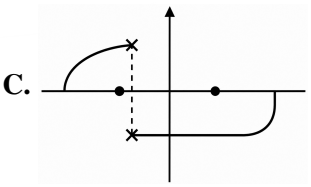
Find the value of K (if it exists) so that the damping ξ of the complex closed loop poles is 0.3.

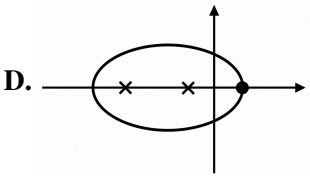
Q5.92.1 MSQ 1992 1M

Which of the following figure(s) represent valid root loci in the s-plane for positive K? Assume that the system has transfer function with real coefficient.

A.


B.


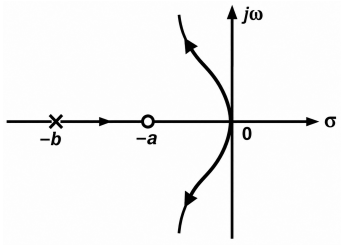
C.


D.


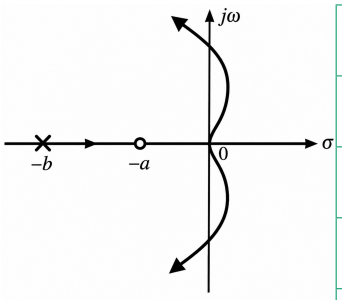
Q5.91.1 MCQ 1991 1M

A unity feedback system has an open loop transfer function of the form $KG(s) = \frac{K(s+a)}{s^2(s+b)}; \quad b > a$, which of the loci shown in the figure can be valid root-loci for the system?

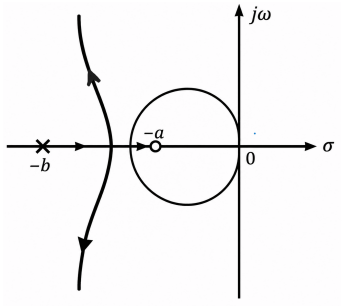
A.



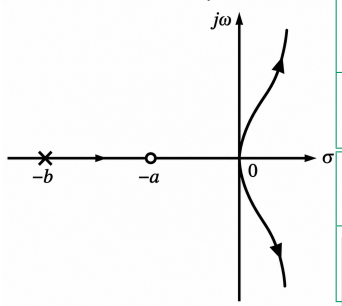
B.



C.



D.



Q5.25.1 A,C	Q5.24.1 4 to 8	Q5.17.1 B
Q5.16.1 A	Q5.15.1 D	Q5.15.2 A
Q5.14.1 C	Q5.11.1 A	Q5.10.1 C
Q5.06.1 B	Q5.05.1 A	Q5.02.1 B
Q5.02.2 *	Q5.01.1 *	Q5.00.1 *
Q5.00.2 *	Q5.00.3 *	Q5.92.1 A
Q5.91.1 A	Q5.91.2 K>2	Q5.91.3 $\pm j\,2\sqrt{2}$
Q5.91.4 *		

Common Data Questions

The next 3 questions share the data below.

Common data for Q5.91.2, Q5.91.3, Q5.91.4
A unity feedback system has the forward loop transfer function $G(s) = \dfrac{K(s+2)^2}{s^2(s-1)}$

Q5.91.2 NAT 1991 5M

Determine the range of K for stable operation.

_____

Q5.91.3 NAT 1991 5M

Determine the imaginary axis crossover points.

_____

Q5.91.4 NAT 1991 5M

Without calculating the real axis break-away points. Sketch the form of root loci for the system.

_____

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

UNIT

VI

Polar Plot and Frequency Response of Second Order Systems

Topics

1. Basics of Frequency Response Analysis
2. Frequency Response of 2nd Order Systems
3. Polar Plot
4. Gain and Phase Crossover Frequency
5. Gain Margin and Phase Margin

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 25	**MCQ:** 15	**MSQ:** 0	**NAT:** 10
1M: 10	**2M:** 7	**Descriptive:** 8	**Avg Weightage** ~2M	**Range:** 0M(2014)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q6.24.1 NAT 2024 2M

Consider the stable closed-loop system shown in the figure. The magnitude and phase values of the frequency response of $G(s)$ are given in the table. The value of the gain K_I (> 0) for a 50° phase margin is

_____(*rounded off to 2 decimal places*).

ω in rad/sec	Magnitude in dB	Phase in degrees
0.5	−7	−40
1.0	−10	−80
2.0	−18	−130
10.0	−40	−200

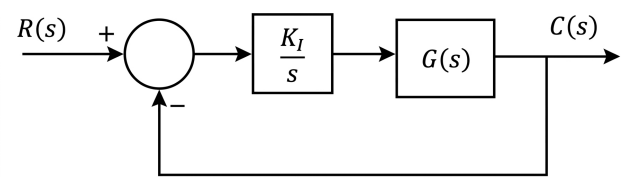


Fig. Q6.24.1

Q6.22.1 MCQ 2022 2M

An LTI system is shown in the figure where

$$G(s) = \frac{100}{s^2 + 0.1s + 10}$$

The steady state output of the system, to the input $r(t)$, is given as $y(t) = a + b\sin(10t + \theta)$. The values of a and b will be

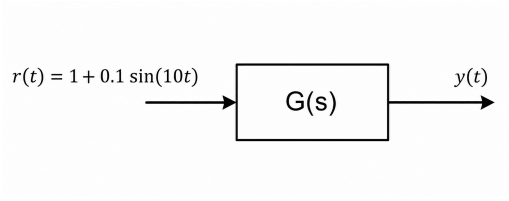


Fig. Q6.22.1

A. $a = 1, b = 10$ **B.** $a = 10, b = 1$
C. $a = 1, b = 100$ **D.** $a = 100, b = 1$

Q6.17.1 MCQ 2017 1M

The transfer function of a system is given by,

$$\frac{V_o(s)}{V_i(s)} = \frac{1 - s}{1 + s}$$

Let the output of the system be $v_o(t) = V_m \sin(\omega t + \phi)$ for the input, $v_i(t) = V_m \sin(\omega t)$. Then the minimum and maximum values of ϕ (in radians) are respectively

A. $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ **B.** $-\frac{\pi}{2}$ and 0

C. 0 and $\frac{\pi}{2}$ **D.** $-\pi$ and 0

Q6.17.2 NAT 2017 1M

Consider the unity feedback control system shown. The value of K that results in a phase margin of the system to be 30° is

_____. (*Give the answer up to two decimal places*)

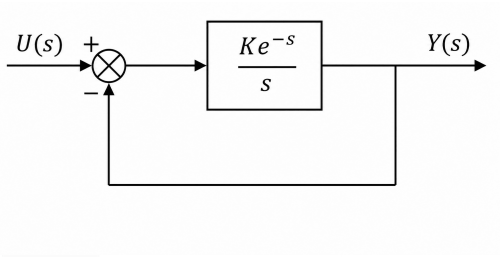


Fig. Q6.17.2

Q6.16.1 MCQ 2016 1M

The transfer function of a system is

$$\frac{Y(s)}{R(s)} = \frac{s}{s + 2}.$$

The steady state output $y(t)$ is $A\cos(2t + \varphi)$ for the input $\cos(2t)$. The values of A and φ, respectively are

A. $\frac{1}{\sqrt{2}}, -45°$ **B.** $\frac{1}{\sqrt{2}}, +45°$

C. $\sqrt{2}, -45°$ **D.** $\sqrt{2}, +45°$

Q6.16.2 MCQ 2016 1M

The phase cross-over frequency of the transfer function $G(s) = \frac{100}{(s + 1)^3}$ in rad/s is

A. $\sqrt{3}$ **B.** $\frac{1}{\sqrt{3}}$ **C.** 3 **D.** $3\sqrt{3}$

Q6.15.1 MCQ 2015 2M

The transfer function of a second order real system with a perfectly flat magnitude response of unity has a pole at $(2 - j3)$. List all the poles and zeroes.

A. Poles at $(2 \pm j3)$, no zeroes.

B. Poles at $(\pm 2 - j3)$, one zero at origin.

C. Poles at $(2 - j3)$, $(-2 + j3)$, zeroes at $(-2 - j3)$, $(2 + j3)$.

D. Poles at $(2 \pm j3)$, zeroes at $(-2 \pm j3)$.

Q6.14.1 MCQ 2014 1M

Consider an LTI system with transfer function

$$H(s) = \frac{1}{s(s+4)}$$

If the input to the system is $\cos(3t)$ and the steady state output is $A\sin(3t + \alpha)$, then the value of A is

A. $\frac{1}{30}$ **B.** $\frac{1}{15}$ **C.** $\frac{3}{4}$ **D.** $\frac{4}{3}$

Q6.12.1 MCQ 2012 1M

A system with transfer function

$$G(s) = \frac{(s^2+9)(s+2)}{(s+1)(s+3)(s+4)}$$

is excited by $\sin(\omega t)$. The steady-state output of the system is zero at

A. $\omega = 1$ rad/s **B.** $\omega = 2$ rad/s
C. $\omega = 3$ rad/s **D.** $\omega = 4$ rad/s

Q6.11.1 MCQ 2011 1M

The frequency response of a linear system $G(j\omega)$ is provided in the tabular form below

$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
1.3	$-130°$
1.2	$-140°$
1.0	$-150°$
0.8	$-160°$
0.5	$-180°$
0.3	$-200°$

The gain margin and phase margin of the system are

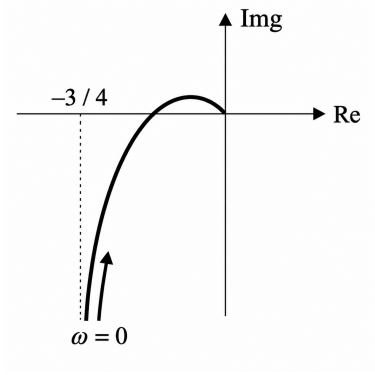
A. 6 dB and 30° **B.** 6 dB and $-30°$
C. -6 dB and 30° **D.** -6 dB and $-30°$

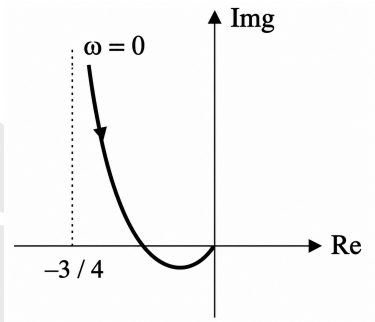
Q6.10.1 MCQ 2010 2M

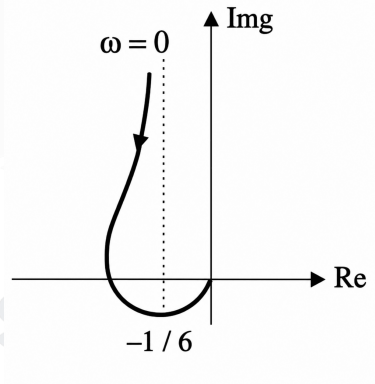
The frequency response of

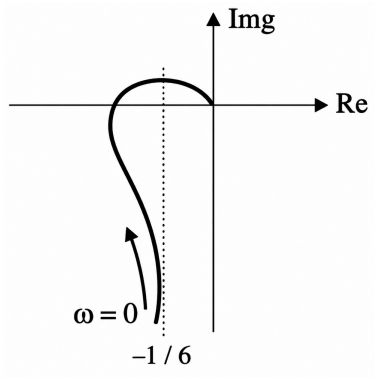
$$G(s) = \frac{1}{s(s+1)(s+2)}$$

plotted in the complex $G(j\omega)$ plane (for $0 < \omega < \infty$) is

A.


B.


C.


D.


Q6.09.1 MCQ 2009 2M

The open loop transfer function of a unity feedback system is given by

$$G(s) = \frac{e^{-0.1s}}{s}$$

.The gain margin of this system is

A. 11.95 dB **B.** 17.67 dB **C.** 21.33 dB **D.** 23.9 dB

Q6.05.1 MCQ 2005 1M

The gain margin of a unity feedback control system with the open loop transfer function

$$G(s) = \frac{(s+1)}{s^2}$$

is:

A. 0 **B.** $\frac{1}{\sqrt{2}}$ **C.** $\sqrt{2}$ **D.** ∞

Q6.04.1 MCQ 2004 2M

In the system shown in figure, the input $x(t) = \sin t$. In the steady-state, the response $y(t)$ will be

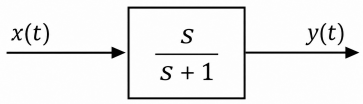


Fig. Q6.04.1

A. $\frac{1}{\sqrt{2}}\sin(t-45°)$ **B.** $\frac{1}{\sqrt{2}}\sin(t+45°)$

C. $\sin(t-45°)$ **D.** $\sin(t+45°)$

Q6.04.2 MCQ 2004 2M

The open loop transfer function of a unity feedback control system is given as $G(s) = \frac{as+1}{s^2}$. The value of 'a' to give a phase margin of 45° is equal to

A. 0.141 **B.** 0.441 **C.** 0.841 **D.** 1.141

Common Data Questions

The next 4 questions share the data below.

Answer Questions Q6.01.1, Q6.01.2, Q6.01.3 and Q6.01.4 using the following data. These four questions together carry a total of **5 marks**.

A unity feedback system has an open-loop transfer function of

$$G(s) = \frac{10000}{s(s+10)^2}$$

Q6.01.1 NAT 2001

Determine the magnitude of $G(j\omega)$ in dB at an angular frequency of $\omega = 20$ rad/sec.

Q6.01.2 NAT 2001

Determine the phase margin in degrees.

Q6.01.3 NAT 2001

Determine the gain margin in dB.

Q6.01.4 NAT 2001

Is the system stable or unstable? (0 - unstable and 1 - stable)

Common Data Questions

Answer Questions Q6.00.1, Q6.00.2 and Q6.00.3 using the following data. These three questions together carry a total of **5 marks**.

Open-loop transfer function of a unity-feedback system is:

$$G(s) = G_1(s)e^{-s\tau_D} = \frac{e^{-s\tau_D}}{s(s+1)(s+2)}$$

Given: $|G_1(j\omega)| = 1$ when $\omega = 0.466$.

Q6.00.1 NAT 2000

Determine the phase margin (in degrees) when $\tau_D = 0$.

Q6.00.2 NAT 2000

Comment in one sentence on the effect of dead time on the stability of the system.

_____

Q6.00.3 NAT 2000

Determine the maximum value of dead time τ_D for the closed-loop system to be stable.

_____

Q6.97.1 NAT 1997 1M

A unity feedback system with the open loop transfer function $G(s) = \frac{1}{s(s+2)(s+4)}$ has gain margin of _____ dB.

Q6.96.1 MCQ 1996 1M

The closed-loop transfer function of a control system is given by

$$\frac{C(s)}{R(s)} = \frac{1}{1+s}.$$

For the input $r(t) = \sin t$, the steady-state value of $c(t)$ is equal to

A. $\frac{1}{\sqrt{2}}\cos t$

B. 1

C. $\frac{1}{\sqrt{2}}\sin t$

D. $\frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right)$

Q6.94.1 MCQ 1994 4M

Match the polar plot for following open loop transfer functions

a. $\frac{s}{(s+1)(s+2)}$ b. $\frac{s^2+1}{s^3}$ c. $\frac{s^2-1}{s^2+1}$

d. $\frac{1}{s^2+10}$

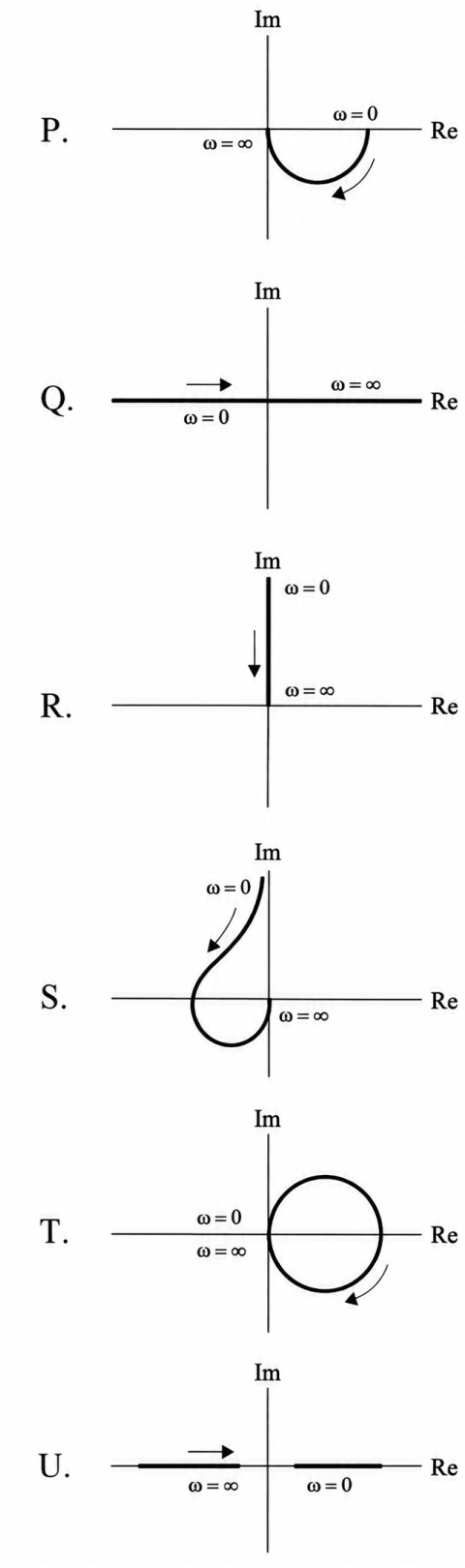


Fig. Q6.94.1

A. $a \to T,\ b \to Q,\ c \to S,\ d \to R$

B. $a \rightarrow T,\ b \rightarrow R,\ c \rightarrow Q,\ d \rightarrow U$

C. $a \rightarrow P,\ b \rightarrow S,\ c \rightarrow R,\ d \rightarrow U$

D. $a \rightarrow T,\ b \rightarrow P,\ c \rightarrow R,\ d \rightarrow U$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q6.24.1 **1.11 to 1.13**	Q6.22.1 **A**	Q6.17.1 **D**
Q6.17.2 **1.01 to 1.06**	Q6.16.1 **B**	Q6.16.2 **A**
Q6.15.1 **D**	Q6.14.1 **B**	Q6.12.1 **C**
Q6.11.1 **A**	Q6.10.1 **A**	Q6.09.1 **D**
Q6.05.1 **D**	Q6.04.1 **B**	Q6.04.2 **C**
Q6.01.1 **0**	Q6.01.2 −36.87	Q6.01.3 −13.98
Q6.01.4 0	Q6.00.1 51.91	Q6.00.2 *
Q6.00.3 1.94	Q6.97.1 **33.62**	Q6.96.1 **D**
Q6.94.1 **B**		

PrepFusion

UNIT

VII

Nyquist Plot

Topics

1. Nyquist Plot
2. Nyquist Stability Criterion
3. Gain Crossover Frequency (ω_{gc}) from Nyquist Plot
4. Phase Crossover Frequency (ω_{pc}) from Nyquist Plot
5. Gain Margin (GM) using Nyquist Plot
6. Phase Margin (PM) using Nyquist Plot

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 17	**MCQ:** 13	**MSQ:** 0	**NAT:** 4
1M: 8	**2M:** 5	**Descriptive:** 4	**Avg Weightage** ~2M	**Range:** 1M(2025)–2M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q7.25.1 MCQ 2025 1M

The Nyquist plot of a strictly stable $G(s)$ having the numerator polynomial as $(s-3)$ encircles the critical point -1 once in the anti-clockwise direction. Which one of the following statements on the closed-loop system shown in figure, is correct?

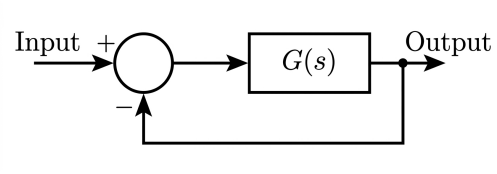


Fig. Q7.25.1

A. The system stability cannot be ascertained.

B. The system is marginally stable.

C. The system is stable.

D. The system is unstable.

Q7.23.1 MCQ 2023 1M

In the Nyquist plot of the open-loop transfer function

$$G(s)H(s) = \frac{3s+5}{s-1}$$

corresponding to the feedback loop shown in the figure, the infinite semi-circular arc of the Nyquist contour in s-plane is mapped into a point at

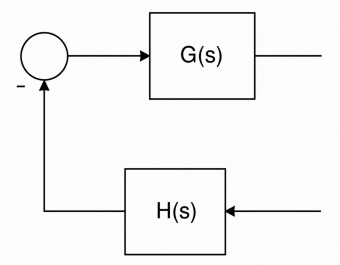


Fig. Q7.23.1

A. $G(s)H(s) = \infty$ **B.** $G(s)H(s) = 0$

C. $G(s)H(s) = 3$ **D.** $G(s)H(s) = -5$

Q7.22.1 MCQ 2022 2M

The open loop transfer function of a unity gain negative feedback system is given as

$$G(s) = \frac{1}{s(s+1)}$$

The Nyquist contour in the s-plane encloses the entire right half plane and a small neighbourhood around the origin in the left half plane, as shown in the figure below.

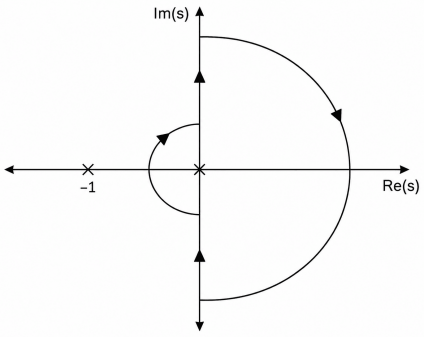


Fig. Q7.22.1

The number of encirclements of the point $(-1+j0)$ by the Nyquist plot of $G(s)$, corresponding to the Nyquist contour, is denoted as N. Then N equals to

A. 0 **B.** 1 **C.** 2 **D.** 3

Q7.20.1 NAT 2020 1M

Consider a negative unity feedback system with forward path transfer function $G(s) = \dfrac{K}{(s+a)(s-b)(s+c)}$, where K, a, b, c are positive real numbers. For a Nyquist path enclosing the entire imaginary axis and right half of the s-plane in the clockwise direction, the Nyquist plot of $(1+G(s))$, encircles the origin of $(1+G(s))$-plane once in the clockwise direction and never passes through this origin for a certain value of K. Then, the number of poles of $\dfrac{G(s)}{1+G(s)}$ lying in the open right half of the s-plane is _____

Q7.19.1 MCQ 2019 1M

The open loop transfer function of a unity feedback system is given by $G(s) = \dfrac{\pi e^{-0.25s}}{s}$. In $G(s)$ plane, the Nyquist plot of $G(s)$ passes through the negative real axis at the point

A. $(-0.5, j0)$ **B.** $(-0.75, j0)$
C. $(-1.25, j0)$ **D.** $(-1.5, j0)$

Q7.16.1 MCQ 2016 2M

Loop transfer function of a feedback system is

$$G(s)H(s) = \frac{s+3}{s^2(s-3)}.$$

Take the Nyquist contour in the clockwise direction. Then, the Nyquist plot of $G(s)H(s)$ encircles $-1 + j0$

A. once in clockwise direction

B. twice in clockwise direction

C. once in anticlockwise direction

D. twice in anticlockwise direction

Q7.15.1 MCQ 2015 1M

Nyquist plots of two functions $G_1(s)$ and $G_2(s)$ are shown in figure.

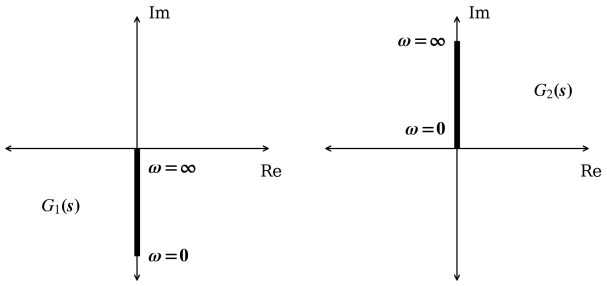
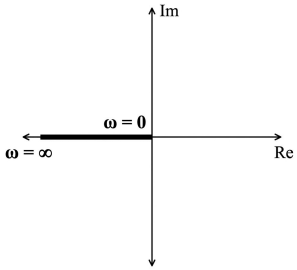


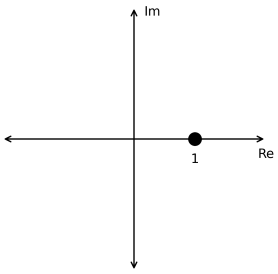
Fig. Q7.15.1

Nyquist plot of the product of $G_1(s)$ and $G_2(s)$ is

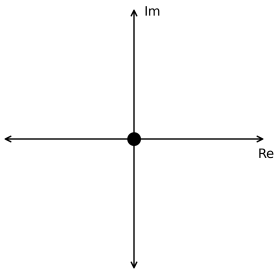
A.



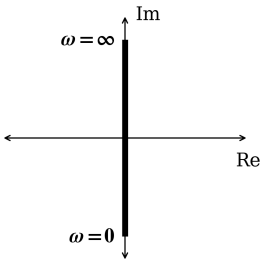
B.



C.



D.



Q7.09.1 MCQ 2009 1M

The polar plot of an open loop stable system is shown below. The closed loop system is

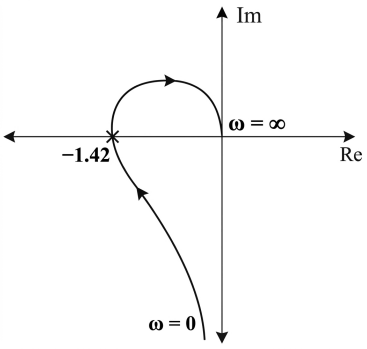


Fig. Q7.09.1

A. always stable

B. marginally stable

C. unstable with one pole on the RH s-plane

D. unstable with two poles on the RH s-plane

Q7.07.1 MCQ 2007 2M

If $x = \Re\,(G(j\omega))$, and $y = \Im\,(G(j\omega))$ then for $\omega \to 0^+$, the Nyquist plot for

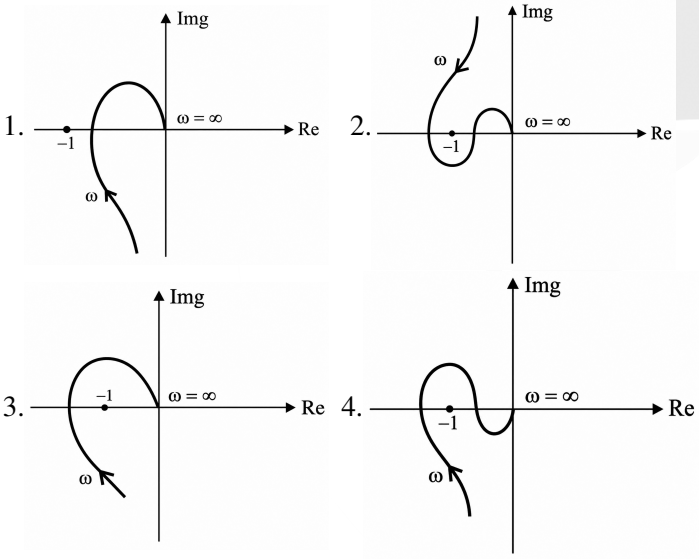
$$G(s) = \frac{1}{s(s+1)(s+2)}$$

becomes asymptotic to the line

A. $x = 0$ **B.** $x = -\frac{3}{4}$ **C.** $x = y - \frac{1}{6}$ **D.** $x = \frac{y}{\sqrt{3}}$

Q7.06.1 MCQ 2006 2M

Consider the following Nyquist plots of loop transfer functions over $\omega = 0$ to $\omega = \infty$. Which of these plots represents a stable closed loop system?



A. (1) only **B.** all, except (1)
C. all, except (3) **D.** (1) and (2) only

Q7.05.1 MCQ 2005 2M

In the $GH(s)$ plane, the Nyquist plot of the loop transfer function

$$G(s)H(s) = \frac{\pi e^{-0.25s}}{s}$$

passes through the negative real axis at the point

A. $(-0.25, j0)$ **B.** $(-0.5, j0)$
C. $(-1, j0)$ **D.** $(-2, j0)$

Q7.04.1 MCQ 2004 1M

The Nyquist plot of loop transfer function $G(s)H(s)$ of a closed loop control system passes through the point $(-1, j0)$ in the $G(s)H(s)$ plane. The phase margin of the system is

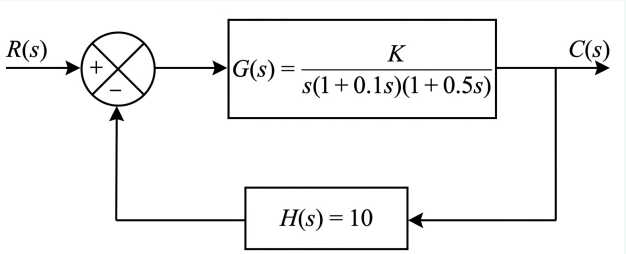
A. 0° **B.** 45° **C.** 90° **D.** 180°

Common Data Questions

The next 3 questions share the data below.

Answer Questions Q7.95.1, Q7.95.2 and Q7.95.3 using the following data. These three questions together carry a total of **5 marks**.

The block diagram of control system is given in figure.



Q7.95.1 NAT 1995

Sketch the Nyquist locus for $0 < \omega < \infty$.

Q7.95.2 NAT 1995

Find gain margin and phase crossover frequency.

Q7.95.3 NAT 1995

Find phase margin (in degrees) of the system for $K = 1.2$.

Q7.92.1 MCQ 1992 1M

A unity feedback system has the open loop transfer function

$$G(s) = \frac{1}{(s-1)(s+2)(s+3)}$$

The Nyquist plot of G encircles the origin

A. Never **B.** Once **C.** Twice **D.** Thrice

Q7.91.1 MCQ 1991

Which of the following is the transfer function of a system having the Nyquist plot shown in figure below?

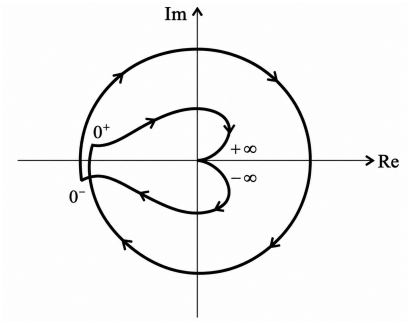


Fig. Q7.91.1

A. $\dfrac{K}{s(s+2)^2(s+5)}$

B. $\dfrac{K}{s^2(s+2)(s+5)}$

C. $\dfrac{K(s+1)}{s^2(s+2)(s+5)}$

D. $\dfrac{K(s+1)(s+3)}{s^2(s+2)(s+5)}$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q7.25.1 D	Q7.23.1 C	Q7.22.1 B
Q7.20.1 2 to 2	Q7.19.1 A	Q7.16.1 A
Q7.15.1 B	Q7.09.1 D	Q7.07.1 B
Q7.06.1 A	Q7.05.1 B	Q7.04.1 A
Q7.95.1 *	Q7.95.2 *	Q7.95.3 0
Q7.92.1 B	Q7.91.1 B	

UNIT

VIII

Bode Plot

Topics

1. Bode Gain and Phase Plots
2. Minimum and Non-Minimum Phase Systems
3. Bode Plots for Repeated Poles and Zeros
4. Recovery of Transfer Function from Bode Plot
5. Error in Asymptotic Bode Gain Plot
6. Bode Plots of 2nd Order Systems

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 19	**MCQ:** 15	**MSQ:** 1	**NAT:** 3
1M: 6	**2M:** 12	**Descriptive:** 1	**Avg Weightage** ~2M	**Range:** 1M(2026)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q8.26.1 MCQ 2026 1M

The asymptotic Bode magnitude plot of a system is shown.

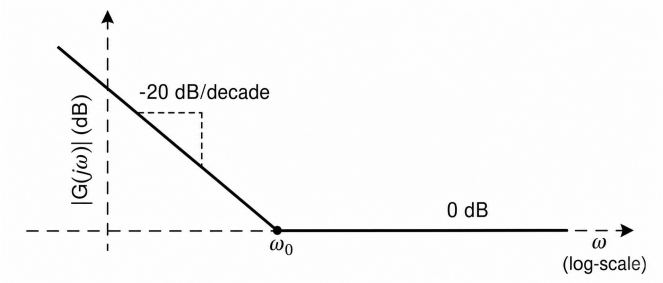


Fig. Q8.26.1

Which one of the following options best represents the transfer function of the system?

A. $G(s) = \dfrac{1 + \dfrac{s}{\omega_0}}{\dfrac{s}{\omega_0}}$

B. $G(s) = \dfrac{\dfrac{s}{\omega_0}}{1 + \dfrac{s}{\omega_0}}$

C. $G(s) = \dfrac{1 + \dfrac{s}{\omega_0}}{1 - \dfrac{s}{\omega_0}}$

D. $G(s) = \dfrac{1 - \dfrac{s}{\omega_0}}{1 + \dfrac{s}{\omega_0}}$

Q8.24.2 NAT 2024 2M

Consider the stable closed-loop system shown in the figure. The asymptotic Bode magnitude plot of $G(s)$ has a constant slope of -20 dB/decade at least till 100 rad/sec with the gain crossover frequency being 10 rad/sec. The asymptotic Bode phase plot remains constant at $-90°$ at least till $\omega =$ 10 rad/sec. The steady-state error of the closed-loop system for a unit ramp input is

_____ (*rounded off to* 2 *decimal places*).

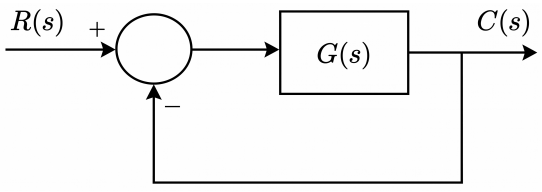


Fig. Q8.24.2

Q8.23.1 MCQ 2023 2M

The asymptotic Bode magnitude and phase plots of a transfer function $G(s)$ are shown in the figure.

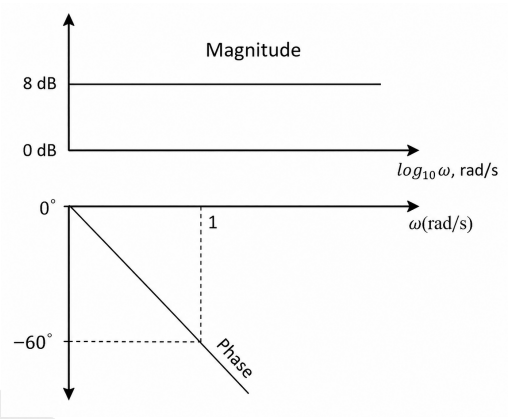


Fig. Q8.23.1

The transfer function $G(s)$ is

A. $2.51e^{-0.032s}$

B. $\dfrac{e^{-2.514s}}{s+1}$

C. $1.04e^{-2.514s}$

D. $2.51e^{-1.047s}$

Q8.22.1 MCQ 2022 1M

The Bode magnitude plot of a first order stable system is constant with frequency. The asymptotic value of the high frequency phase, for the system, is $-180°$. This system has

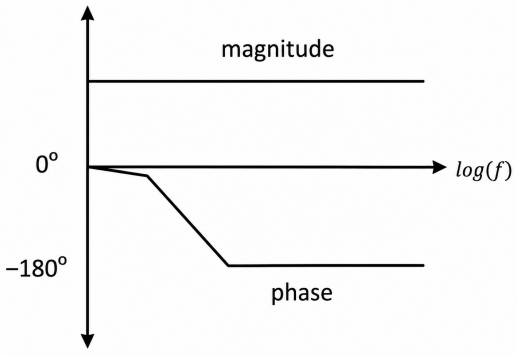


Fig. Q8.22.1

A. one LHP pole and one RHP zero at the same frequency.

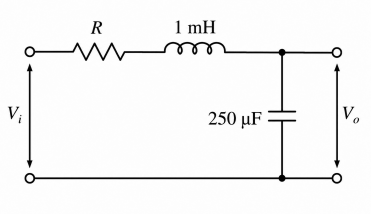
B. one LHP pole and one LHP zero at the same frequency.

C. two LHP poles and one RHP zero.

D. two RHP poles and one LHP zero.

Q8.21.1 NAT 2021 1M

The Bode magnitude plot for the transfer function $\frac{V_o(s)}{V_i(s)}$ of the circuit is as shown. The value of R is _____ Ω. (*Round off to 2 decimal places*)



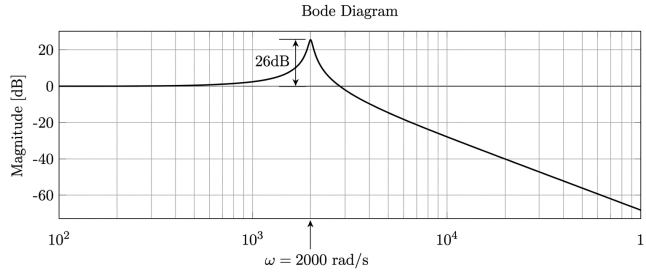


Fig. Q8.21.1

Q8.19.1 MCQ 2019 2M

The asymptotic Bode magnitude plot of a minimum phase transfer function $G(s)$ is shown below.

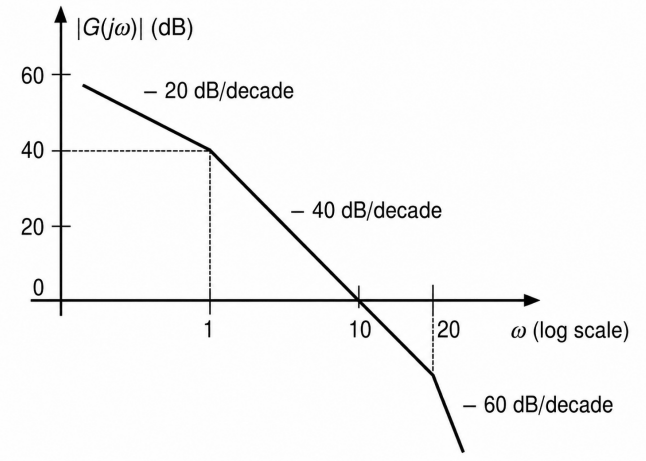


Fig. Q8.19.1

Consider the following two statements.
Statement I: Transfer function $G(s)$ has three poles and one zero.
Statement II: At very high frequency ($\omega \to \infty$), the phase angle $\angle G(j\omega) = -\frac{3\pi}{2}$.

Which one of the following options is correct?

A. Statement I is true and statement II is false.

B. Statement I is false and statement II is true.

C. Both the statements are true.

D. Both the statements are false.

Q8.16.1 MCQ 2016 2M

Consider the following asymptotic Bode magnitude plot (ω is in rad/s).

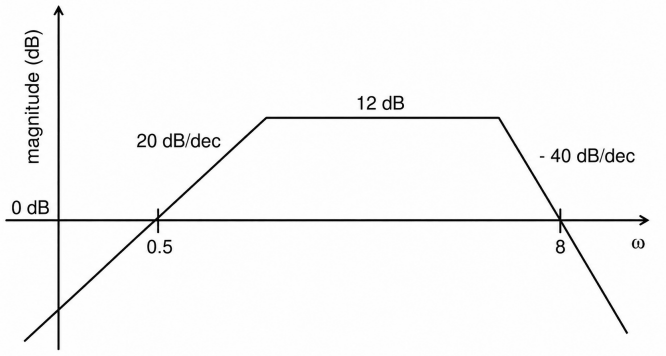


Fig. Q8.16.1

Which one of the following transfer functions is best represented by the above Bode magnitude plot?

A. $\frac{2s}{(1+0.5s)(1+0.25s)^2}$ **B.** $\frac{4(1+0.5s)}{s(1+0.25s)}$

C. $\frac{2s}{(1+2s)(1+4s)}$ **D.** $\frac{4s}{(1+2s)(1+4s)^2}$

Q8.15.1 MCQ 2015 1M

A Bode magnitude plot for the transfer function $G(s)$ of a plant is shown in the figure. Which one of the following transfer functions best describes the plant?

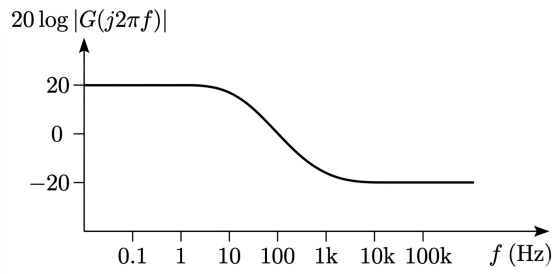


Fig. Q8.15.1

A. $\dfrac{1000(s+10)}{s+1000}$ **B.** $\dfrac{10(s+10)}{s(s+1000)}$

C. $\dfrac{s+1000}{10s(s+10)}$ **D.** $\dfrac{s+1000}{10(s+10)}$

Q8.14.1 NAT 2014 2M

The Bode magnitude plot of the transfer function

$$G(s)=\frac{K(1+0.5s)(1+as)}{s\left(1+\frac{s}{8}\right)(1+bs)\left(1+\frac{s}{36}\right)}$$

is shown below:
(*Note that* $-6\,dB/octave = -20\,dB/decade$)

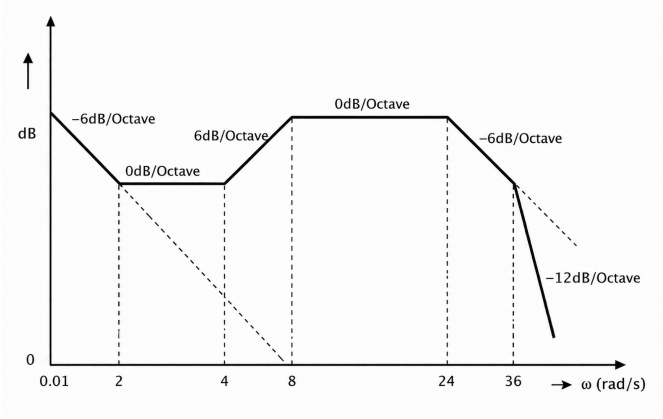


Fig. Q8.14.1

The value of $\dfrac{a}{bK}$ is

Q8.14.2 MCQ 2014 2M

In the figure shown, assume the op-amp to be ideal. Which of the alternatives gives the correct Bode plots for the transfer function $\dfrac{V_o(\omega)}{V_i(\omega)}$?

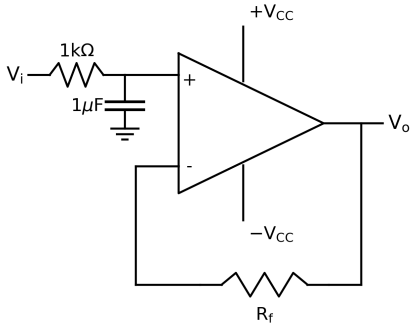
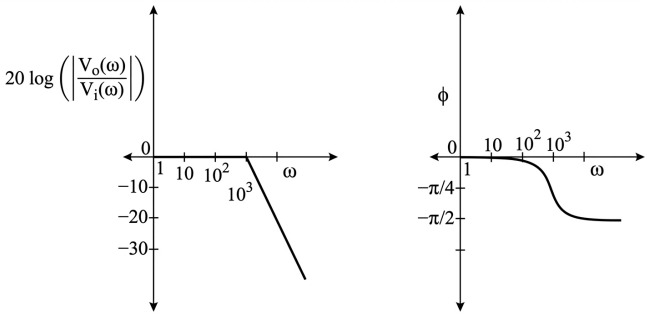
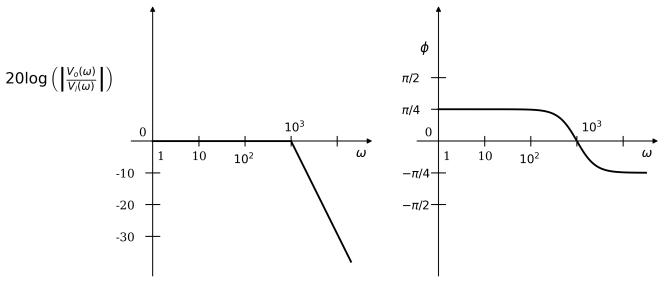


Fig. Q8.14.2

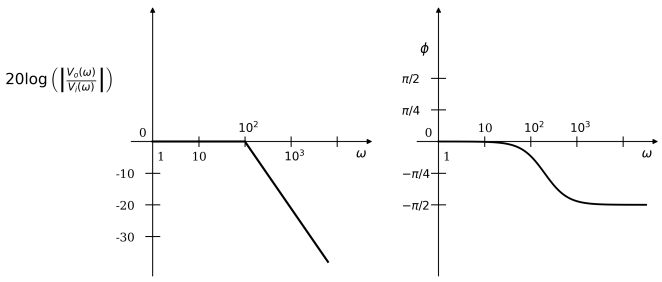
A.



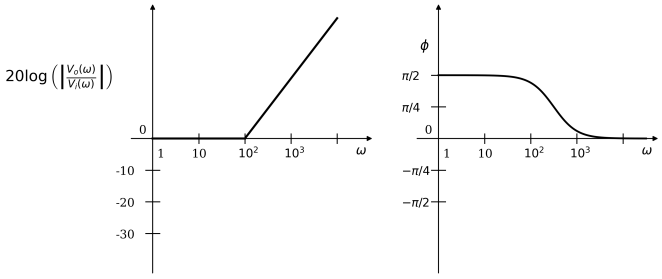
B.



C.



D.



Q8.14.3 MCQ 2014 2M

For the transfer function

$$G(s) = \frac{5(s+4)}{s(s+0.25)(s^2+4s+25)}$$

The values of the constant gain term and the highest corner frequency of the Bode plot respectively are

A. 3.2, 5.0 **B.** 16.0, 4.0 **C.** 3.2, 4.0 **D.** 16.0, 5.0

Q8.13.1 MCQ 2013 1M

The Bode plot of a transfer function $G(s)$ is shown in the figure below.

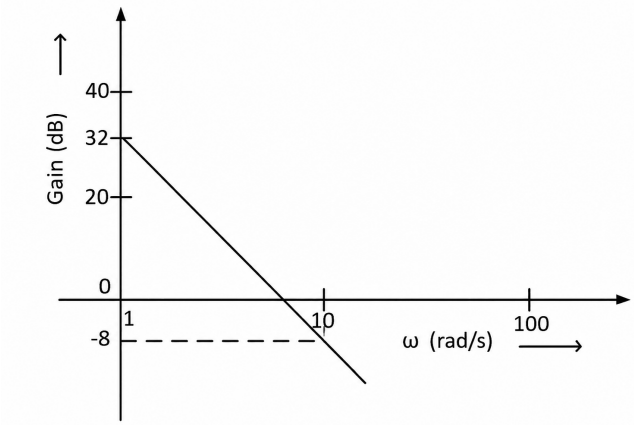


Fig. Q8.13.1

The gain $(20\log|G(s)|)$ is 32 dB and −8 dB at 1 rad/s and 10 rad/s respectively. The phase is negative for all ω. Then $G(s)$ is

A. $\frac{39.8}{s}$ **B.** $\frac{39.8}{s^2}$ **C.** $\frac{32}{s}$ **D.** $\frac{32}{s^2}$

Q8.09.1 MCQ 2009 1M

The asymptotic approximation of the log-magnitude vs frequency plot of a system containing only real poles and zeros is shown. Its transfer function is

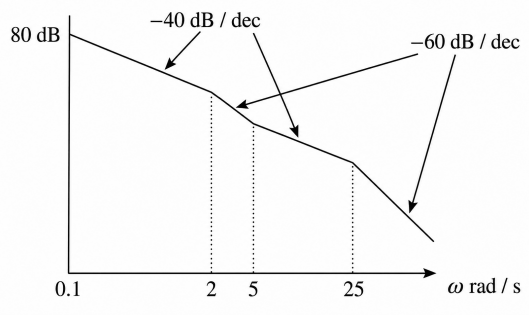


Fig. Q8.09.1

A. $\frac{10(s+5)}{s(s+2)(s+25)}$ **B.** $\frac{1000(s+5)}{s^2(s+2)(s+25)}$

C. $\frac{100(s+5)}{s(s+2)(s+25)}$ **D.** $\frac{80(s+5)}{s^2(s+2)(s+25)}$

Q8.08.1 MCQ 2008 2M

The asymptotic Bode magnitude plot of a minimum phase transfer function is shown in the figure:

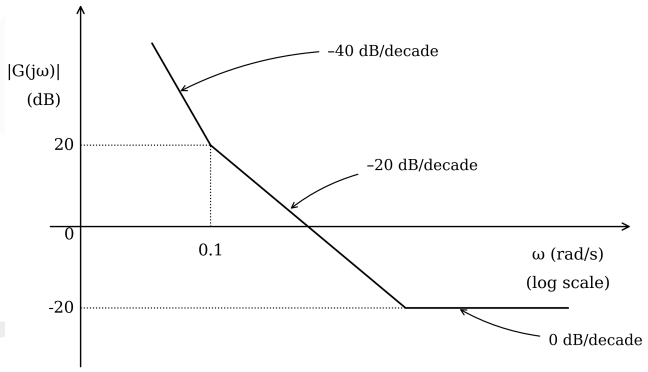


Fig. Q8.08.1

This transfer function has

A. Three poles and one zero

B. Two poles and one zero

C. Two poles and two zeros

D. One pole and two zeros

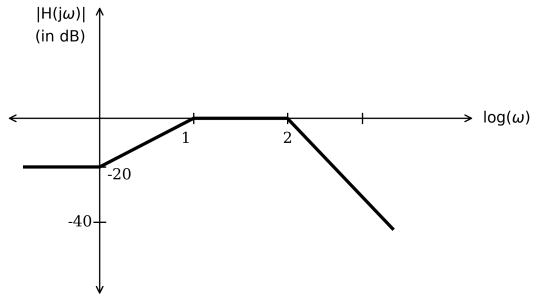
Q8.06.1 MCQ 2006 2M

The Bode magnitude plot of the transfer function

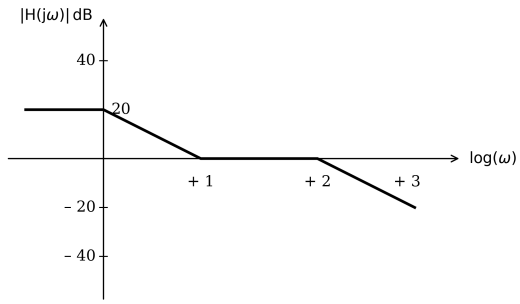
$$H(j\omega) = \frac{10^4(1+j\omega)}{(10+j\omega)(100+j\omega)^2}$$

is

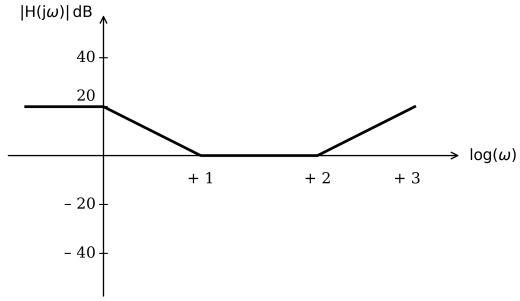
A.



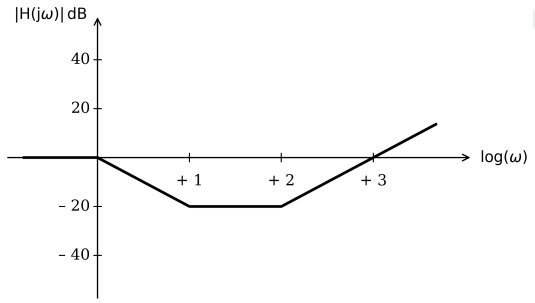
B.



C.



D.



Q8.03.1 MCQ 2003 2M

The asymptotic Bode plot of the transfer function

$$\frac{K}{1+\frac{s}{a}}$$

is given in below figure. The error in phase angle and dB gain at a frequency of $\omega = 0.5\,a$ are respectively.

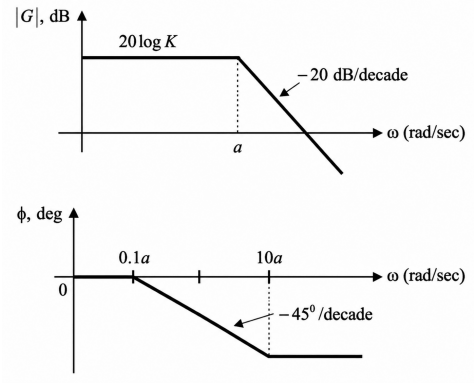


Fig. Q8.03.1

A. 4.9°, 0.97 dB
B. 5.7°, 3 dB
C. 4.9°, 3 dB
D. 5.7°, 0.97 dB

Q8.01.1 MCQ 2001 2M

The asymptotic approximation of the log-magnitude versus frequency plot of a minimum phase system with real poles and one zero is shown in the below figure. Its transfer function is

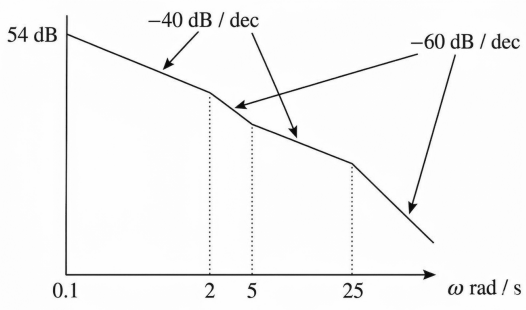


Fig. Q8.01.1

A. $\dfrac{20(s+5)}{s(s+2)(s+25)}$
B. $\dfrac{10(s+5)}{(s+2)^2(s+25)}$
C. $\dfrac{20(s+5)}{s^2(s+2)(s+25)}$
D. $\dfrac{50(s+5)}{s^2(s+2)(s+25)}$

Q8.99.1 MCQ 1999 2M

The function corresponding to the Bode plot is:

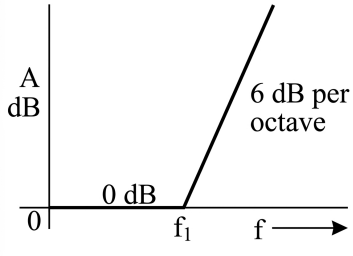


Fig. Q8.99.1

A. $A = jf/f_1$

B. $A = \dfrac{1}{(1 - jf_1/f)}$

C. $A = \dfrac{1}{(1 + jf_1/f)}$

D. $A = 1 + jf/f_1$

Q8.91.1 MSQ 1991

The system having the Bode magnitude plot shown in figure below has the transfer function

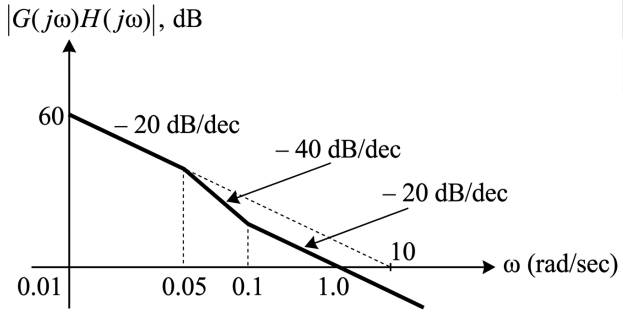


Fig. Q8.91.1

☐ **A.** $\dfrac{60(s + 0.01)(s + 0.1)}{s^2(s + 0.05)^2}$

☐ **B.** $\dfrac{10(1 + 10s)}{s(1 + 20s)}$

☐ **C.** $\dfrac{3(s + 0.05)}{s(s + 0.1)(s + 1)}$

☐ **D.** $\dfrac{5(s + 0.1)}{s(s + 0.05)}$

Answer Key

Q8.26.1 **A**	Q8.24.2 **0.09 to 0.11**	Q8.23.1 **D**
Q8.22.1 **A**	Q8.21.1 **0.09 to 0.11**	Q8.19.1 **B**
Q8.16.1 **A**	Q8.15.1 **D**	Q8.14.1 **0.7 to 0.8**
Q8.14.2 **A**	Q8.14.3 **A**	Q8.13.1 **B**
Q8.09.1 **B**	Q8.08.1 **C**	Q8.06.1 **A**
Q8.03.1 **A**	Q8.01.1 **D**	Q8.99.1 **D**
Q8.91.1 **B, D**		

UNIT

IX

State Space Analysis

Topics

1. State Variables and State Diagram
2. State Space Representation
3. Eigen Values and Eigen Vectors in State Space
4. Solution of State Space Equations
5. State Transition Matrix and Its Properties
6. Recovery of Transfer Function from State Space
7. Similarity Transforms
8. Controllable and Observable Canonical Forms
9. Parallel and Cascade Decomposition
10. Controllability — Kalman's and Gilbert's Test
11. Observability — Kalman's and Gilbert's Test
12. Observable and Detectable Systems

Unit Snapshot*

Stream: EE	**Questions:** 47	**MCQ:** 34	**MSQ:** 0	**NAT:** 13
1M: 7	**2M:** 29	**Descriptive:** 11	**Avg Weightage** ~3M	**Range:** 1M(2016)–4M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q9.26.1 MCQ 2026 2M

A system is characterized by the following state equation and output equation (u: input, $\mathbf{x}$: state vector, y: output)

$$\dot{\mathbf{x}} = \begin{bmatrix} a & b \\ -a & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 2 \end{bmatrix} \mathbf{x}$$

What are the values of a and b for which the poles of the transfer function are at $-2 + j3$ and $-2 - j3$?

A. $a = 4,\ b = 3.25$ **B.** $a = -4,\ b = 3.25$
C. $a = 4,\ b = -3.25$ **D.** $a = -4,\ b = -3.25$

Q9.26.2 MCQ 2026 2M

A system is represented in state-space form as follows: (u: input, $\mathbf{x}$: state vector, y: output)

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 2 \end{bmatrix} \mathbf{x}$$

Consider the new state vector

$$\mathbf{z} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x}$$

What is the state-space representation of the system in terms of the new state vector $\mathbf{z}$?

A. $\dot{\mathbf{z}} = \begin{bmatrix} -1 & 4 \\ -1 & -2 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix} u$

$y = \begin{bmatrix} 2 & 3 \end{bmatrix} \mathbf{z}$

B. $\dot{\mathbf{z}} = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 3 \\ 5 \end{bmatrix} u$

$y = \begin{bmatrix} 2 & 3 \end{bmatrix} \mathbf{z}$

C. $\dot{\mathbf{z}} = \begin{bmatrix} 4 & 9 \\ -2 & -3 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix} u$

$y = \begin{bmatrix} 2 & 3 \end{bmatrix} \mathbf{z}$

D. $\dot{\mathbf{z}} = \begin{bmatrix} 2 & 1 \\ -4 & 1 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix} u$

$y = \begin{bmatrix} 4 & -1 \end{bmatrix} \mathbf{z}$

Q9.25.3 NAT 2025 2M

Consider the state-space model

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + Br(t),$$

$$y(t) = C\mathbf{x}(t)$$

where $\mathbf{x}(t)$, $r(t)$, $y(t)$ are the state, input and output, respectively. The matrices A, B, C are given below

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

The sum of the magnitudes of the poles is

_____ (*round off to nearest integer*).

Q9.23.1 NAT 2023 2M

Consider the state-space description of an LTI system with matrices

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & -2 \end{bmatrix}, \quad D = 1$$

For the input, $\sin(\omega t)$, $\omega > 0$, the value of ω for which the steady-state output of the system will be zero, is

_____(*Round off to the nearest integer*).

Q9.21.1 NAT 2021 2M

The state space representation of a first-order system is given as

$$\dot{x} = -x + u$$

$$y = x$$

where, x is the state variable, u is the control input and y is the controlled output. Let $u = -Kx$ be the control law, where K is the controller gain. To place a closed-loop pole at -2, the value of K is

_____.

Q9.19.1 MCQ 2019 2M

Consider a state-variable model of a system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\alpha & -2\beta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \alpha \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

where y is the output, and r is the input. The damping ratio ξ and the undamped natural frequency ω_n (rad/sec) of the system are given by

A. $\xi = \dfrac{\beta}{\sqrt{\alpha}};\ \omega_n = \sqrt{\alpha}$ **B.** $\xi = \sqrt{\alpha};\ \omega_n = \dfrac{\beta}{\sqrt{\alpha}}$

C. $\xi = \dfrac{\sqrt{\alpha}}{\beta};\ \omega_n = \sqrt{\beta}$ **D.** $\xi = \sqrt{\beta};\ \omega_n = \sqrt{\alpha}$

Q9.18.1 MCQ 2018 2M

Consider a system governed by the following equations

$$\frac{dx_1(t)}{dt} = x_2(t) - x_1(t)$$

$$\frac{dx_2(t)}{dt} = x_1(t) - x_2(t)$$

The initial conditions are such that $x_1(0) < x_2(0) < \infty$. Let $x_{1f} = \lim_{t\to\infty} x_1(t)$ and $x_{2f} = \lim_{t\to\infty} x_2(t)$. Which one of the following is true?

A. $x_{1f} < x_{2f} < \infty$ **B.** $x_{2f} < x_{1f} < \infty$

C. $x_{1f} = x_{2f} < \infty$ **D.** $x_{1f} = x_{2f} = \infty$

Q9.17.1 MCQ 2017 2M

The transfer function of the system $Y(s)/U(s)$ whose state-space equations are given below is:

$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix} = \begin{bmatrix}1 & 2\\ 2 & 0\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}1\\ 2\end{bmatrix}u(t)$$

$$y(t) = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix}.$$

A. $\dfrac{(s+2)}{(s^2-2s-2)}$ **B.** $\dfrac{(s-2)}{(s^2+s-4)}$

C. $\dfrac{(s-4)}{(s^2+s-4)}$ **D.** $\dfrac{(s+4)}{(s^2-s-4)}$

Q9.17.2 NAT 2017 2M

Consider the system described by the following state space representation

$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & -2\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u(t)$$

$$y(t) = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix}$$

If $u(t)$ is a unit step input and

$$\begin{bmatrix}x_1(0)\\ x_2(0)\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix},$$

the value of output $y(t)$ at $t = 1$ sec is

_____(*rounded off to three decimal places*)

Q9.16.1 NAT 2016 2M

Consider the following state-space representation of a linear time-invariant system.

$$\dot{x}(t) = \begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}x(t), \qquad y(t) = c^T x(t), \qquad c = \begin{bmatrix}1\\ 1\end{bmatrix}$$

$$x(0) = \begin{bmatrix}1\\ 1\end{bmatrix}$$

The value of $y(t)$ for $t = \log_e 2$ is

_____

Q9.15.1 MCQ 2015 2M

In the signal flow diagram given in the figure, u_1 and u_2 are possible inputs whereas y_1 and y_2 are possible outputs. When would the SISO system derived from this diagram be controllable and observable?

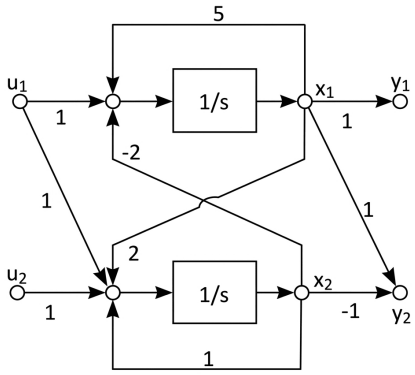


Fig. Q9.15.1

A. When u_1 is the only input and y_1 is the only output.

B. When u_2 is the only input and y_1 is the only output.

C. When u_1 is the only input and y_2 is the only output.

D. When u_2 is the only input and y_2 is the only output.

Q9.15.2 MCQ 2015 2M

For the system governed by the set of equations:

$$\frac{dx_1}{dt} = 2x_1 + x_2 + u$$

$$\frac{dx_2}{dt} = -2x_1 + u$$

$$y = 3x_1$$

the transfer function $Y(s)/U(s)$ is given by

A. $\dfrac{3(s+1)}{s^2-2s+2}$ **B.** $\dfrac{3(2s+1)}{s^2-2s+1}$

C. $\dfrac{s+1}{s^2-2s+1}$ **D.** $\dfrac{3(2s+1)}{s^2-2s+2}$

Q9.14.1 MCQ 2014 1M

The state transition matrix for the system

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}u$$

is

A. $\begin{bmatrix}e^t & 0\\ e^t & e^t\end{bmatrix}$ **B.** $\begin{bmatrix}e^t & 0\\ t^2e^t & e^t\end{bmatrix}$

C. $\begin{bmatrix}e^t & 0\\ te^t & e^t\end{bmatrix}$ **D.** $\begin{bmatrix}e^t & te^t\\ 0 & e^t\end{bmatrix}$

Q9.14.2 MCQ 2014 2M

A discrete system is represented by the difference equation

$$\begin{bmatrix}X_1(k+1)\\ X_2(k+1)\end{bmatrix}=\begin{bmatrix}a & a-1\\ a+1 & a\end{bmatrix}\begin{bmatrix}X_1(k)\\ X_2(k)\end{bmatrix}$$

It has initial conditions $X_1(0)=1;\ X_2(0)=0$. The pole locations of the system for $a=1$, are

A. $1\pm j0$ **B.** $-1\pm j0$ **C.** $\pm1+j0$ **D.** $0\pm j1$

Q9.14.3 MCQ 2014 2M

The second order dynamic system

$$\frac{dX}{dt}=PX+Qu$$

$$y=RX$$

has the matrices P, Q and R as follows:

$$P=\begin{bmatrix}-1 & 1\\ 0 & -3\end{bmatrix}\qquad Q=\begin{bmatrix}0\\ 1\end{bmatrix}\qquad R=\begin{bmatrix}0 & 1\end{bmatrix}$$

The system has the following controllability and observability properties:

A. Controllable and observable

B. Not controllable but observable

C. Controllable but not observable

D. Not controllable and not observable

Q9.14.4 MCQ 2014 2M

Consider the system described by following state space equations

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0 & 1\\ -1 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ 1\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

If u is unit step input, then the steady state error of the system is

A. 0 **B.** $\dfrac{1}{2}$ **C.** $\dfrac{2}{3}$ **D.** 1

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.13.1 and Q9.13.2 using the following data.

The state variable formulation of a system is given as

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}-2 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}u,$$

$$x_1(0)=0,\ x_2(0)=0$$

$$y=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Q9.13.1 MCQ 2013 2M

The system is

A. controllable but not observable

B. not controllable but observable

C. both controllable and observable

D. both not controllable and not observable

Q9.13.2 MCQ 2013 2M

The response $y(t)$ to a unit step input is

A. $\dfrac{1}{2}-\dfrac{1}{2}e^{-2t}$ **B.** $1-\dfrac{1}{2}e^{-2t}-\dfrac{1}{2}e^{-t}$

C. $e^{-2t}-e^{-t}$ **D.** $1-e^{-t}$

Q9.12.1 MCQ 2012 2M

The state variable description of an LTI system is given by

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & a_1 & 0\\ 0 & 0 & a_2\\ a_3 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

where y is the output and u is the input. The system is controllable for

A. $a_1 \neq 0,\ a_2 = 0,\ a_3 \neq 0$ **B.** $a_1 = 0,\ a_2 \neq 0,\ a_3 \neq 0$

C. $a_1 = 0,\ a_2 \neq 0,\ a_3 = 0$ **D.** $a_1 \neq 0,\ a_2 \neq 0,\ a_3 = 0$

Q9.10.1 MCQ 2010 2M

The system $\dot{x} = Ax + Bu$ with

$$A = \begin{bmatrix}-1 & 2\\ 0 & 2\end{bmatrix}, \qquad B = \begin{bmatrix}0\\ 1\end{bmatrix}$$

is

A. stable and controllable

B. stable but uncontrollable

C. unstable but controllable

D. unstable and uncontrollable

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.09.1 and Q9.09.2 using the following data.

A system is described by the following state and output equations

$$\frac{dx_1(t)}{dt} = -3x_1(t) + x_2(t) + 2u(t)$$

$$\frac{dx_2(t)}{dt} = -2x_2(t) + u(t)$$

$$y(t) = x_1(t)$$

where $u(t)$ is the input and $y(t)$ is the output.

Q9.09.1 MCQ 2009 2M

The system transfer function is

A. $\dfrac{s+2}{s^2+5s-6}$ **B.** $\dfrac{s+3}{s^2+5s+6}$

C. $\dfrac{2s+5}{s^2+5s+6}$ **D.** $\dfrac{2s-5}{s^2+5s-6}$

Q9.09.2 MCQ 2009 2M

The state-transition matrix of the above system is

A. $\begin{bmatrix}e^{-3t} & 0\\ e^{-2t}+e^{-3t} & e^{-2t}\end{bmatrix}$ **B.** $\begin{bmatrix}e^{-3t} & e^{-2t}-e^{-3t}\\ 0 & e^{-2t}\end{bmatrix}$

C. $\begin{bmatrix}e^{-3t} & e^{-2t}+e^{-3t}\\ 0 & e^{-2t}\end{bmatrix}$ **D.** $\begin{bmatrix}e^{3t} & e^{-2t}-e^{-3t}\\ 0 & e^{-2t}\end{bmatrix}$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.08.1 and Q9.08.2 using the following data.

The state space equation of a system is described by

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

where x is state vector, u is input, y is output and

$$A = \begin{bmatrix}0 & 1\\ 0 & -2\end{bmatrix}, \qquad B = \begin{bmatrix}0\\ 1\end{bmatrix}, \qquad C = \begin{bmatrix}1 & 0\end{bmatrix}.$$

Q9.08.1 MCQ 2008 2M

The transfer function $G(s)$ of this system will be

A. $\dfrac{s}{(s+2)}$ **B.** $\dfrac{s+1}{s(s-2)}$ **C.** $\dfrac{s}{(s-2)}$ **D.** $\dfrac{1}{s(s+2)}$

Q9.08.2 MCQ 2008 2M

A unity feedback is provided to the above system $G(s)$ to make it a closed loop system as shown in figure.

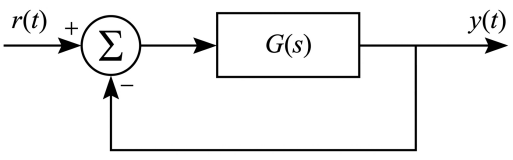


Fig. Q9.08.2

For a unit step input $r(t)$, the steady state error in the output will be

A. 0 **B.** 1 **C.** 2 **D.** ∞

Q9.06.1 MCQ 2006 1M

For a system with the transfer function

$$H(s) = \frac{3(s-2)}{4s^2 - 2s + 1},$$

the matrix A in the state space form $\dot{x} = Ax + Bu$ is equal to

A. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 2 & -4 \end{bmatrix}$ **B.** $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 2 & -4 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 1 & 0 \\ 3 & -2 & 1 \\ 1 & -2 & 4 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 2 & -4 \end{bmatrix}$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.05.1 and Q9.05.2 using the following data.
A state variable system

$$\dot{X}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix} X(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t),$$

with the initial condition

$$X(0) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

and the unit step input $u(t)$ has

Q9.05.1 MCQ 2005 2M

The state transition matrix

A. $\begin{bmatrix} 1 & \frac{1}{3}(1 - e^{-3t}) \\ 0 & e^{-3t} \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & \frac{1}{3}(e^{-t} - e^{-3t}) \\ 0 & e^{-t} \end{bmatrix}$

C. $\begin{bmatrix} 1 & \frac{1}{3}(e^{-t} - e^{-3t}) \\ 0 & e^{-3t} \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & (1 - e^{-t}) \\ 0 & e^{-t} \end{bmatrix}$

Q9.05.2 MCQ 2005 2M

and the state transition equation

A. $X(t) = \begin{bmatrix} t - e^{-t} \\ e^{-t} \end{bmatrix}$ **B.** $X(t) = \begin{bmatrix} t - e^{-t} \\ 3e^{-3t} \end{bmatrix}$

C. $X(t) = \begin{bmatrix} t - e^{-3t} \\ 3e^{-3t} \end{bmatrix}$ **D.** $X(t) = \begin{bmatrix} t - e^{-3t} \\ e^{-t} \end{bmatrix}$

Q9.04.1 MCQ 2004 2M

The state variable description of a linear autonomous system is $\dot{X} = AX$, where X is the two dimensional state vector and A is the system matrix given by

$$A = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}.$$

The roots of the characteristic equation are

A. -2 and $+2$ **B.** $-j2$ and $+j2$
C. -2 and -2 **D.** $+2$ and -2

Q9.03.1 MCQ 2003 1M

A second order system starts with an initial condition of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ without any external input. The state transition matrix for the system is given by

$$\begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-t} \end{bmatrix}.$$

The state of the system at the end of 1 second is given by

A. $\begin{bmatrix} 0.271 \\ 1.100 \end{bmatrix}$ **B.** $\begin{bmatrix} 0.135 \\ 0.368 \end{bmatrix}$ **C.** $\begin{bmatrix} 0.271 \\ 0.736 \end{bmatrix}$ **D.** $\begin{bmatrix} 0.135 \\ 1.100 \end{bmatrix}$

Q9.03.2 MCQ 2003 2M

The following equation defines a separately excited dc motor in the form of

$$\frac{d^2\omega}{dt^2} + \frac{B}{J}\frac{d\omega}{dt} + \frac{K^2}{LJ}\omega = \frac{K}{LJ}V_a$$

The above equation may be organized in the state-space form as follows

$$\begin{bmatrix} \frac{d^2\omega}{dt^2} \\ \frac{d\omega}{dt} \end{bmatrix} = P \begin{bmatrix} \frac{d\omega}{dt} \\ \omega \end{bmatrix} + QV_a$$

where the P matrix is given by

A. $\begin{bmatrix} -\frac{B}{J} & -\frac{K^2}{LJ} \\ 1 & 0 \end{bmatrix}$ **B.** $\begin{bmatrix} -\frac{K^2}{LJ} & -\frac{B}{J} \\ 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 1 \\ \frac{K^2}{LJ} & -\frac{B}{J} \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & 0 \\ -\frac{B}{J} & -\frac{K^2}{LJ} \end{bmatrix}$

Q9.02.1 MCQ 2002 1M

The state transition matrix for the system $\dot{X} = AX$, with initial state $X(0)$ is

A. $(sI - A)^{-1}$

B. $e^{At}X(0)$

C. Laplace inverse of $\left[(sI - A)^{-1}\right]$

D. Laplace inverse of $\left[(sI - A)^{-1}X(0)\right]$

Q9.02.2 MCQ 2002 2M

For the system

$$\dot{X} = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix} X + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u,$$

which of the following statements is true?

A. The system is controllable but unstable

B. The system is uncontrollable but unstable

C. The system is controllable but stable

D. The system is uncontrollable but stable

Q9.02.3 MCQ 2002 2M

For the system

$$\dot{X} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} X + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u; \qquad y = \begin{bmatrix} 4 & 0 \end{bmatrix} X,$$

with u as unit impulse and with zero initial state, the output, y, becomes

A. $2e^{2t}$ **B.** $4e^{2t}$ **C.** $2e^{4t}$ **D.** $4e^{4t}$

Q9.02.4 NAT 2002 5M

Obtain a state variable representation of the system governed by the differential equation

$$\frac{d^2y}{dt^2} + \frac{dy}{dt} - 2y = u(t)e^{-t},$$

with the choice of state variables as

$$x_1 = y, \qquad x_2 = \left(\frac{dy}{dt} - y\right)e^t.$$

Also find $x_2(t)$, given that $u(t)$ is a unit step function and $x_2(0) = 0$.

Q9.01.1 MCQ 2001 1M

Given the homogeneous state-space equation

$$\dot{x} = \begin{bmatrix} -3 & 1 \\ 0 & -2 \end{bmatrix} x$$

the steady state value $x_{ss} = \lim_{t \to \infty} x(t)$, given the initial state value of $x(0) = \begin{bmatrix} 10 & -10 \end{bmatrix}^T$, is

A. $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ **B.** $\begin{bmatrix} -3 \\ -2 \end{bmatrix}$ **C.** $\begin{bmatrix} -10 \\ 10 \end{bmatrix}$ **D.** $\begin{bmatrix} \infty \\ \infty \end{bmatrix}$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.00.1, Q9.00.2 and Q9.00.3 using the following data. These three questions together carry a total of **5 marks**.

Consider the state equation

$$\dot{x}(t) = Ax(t)$$

Given:

$$e^{At} = \begin{bmatrix} e^{-t} + te^{-t} & te^{-t} \\ -te^{-t} & e^{-t} - te^{-t} \end{bmatrix}$$

Q9.00.1 NAT 2000

Find a set of states $x_1(1)$ and $x_2(1)$ such that $x_1(2) = 2$.

Q9.00.2 NAT 2000

Show that

$$(sI - A)^{-1} = \Phi(s) = \frac{1}{\Delta}\begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}, \qquad \Delta = (s+1)^2$$

Q9.00.3 NAT 2000

From $\Phi(s)$, find the matrix A.

Q9.98.1 NAT 1998 5M

The state-space representation of a system is given by:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-5 & 1\\ -6 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Find the Laplace transform of the state transition matrix. Find also the value of x_1 at $t = 1$ if $x_1(0) = 1$ and $x_2(0) = 0$.

Q9.97.1 NAT 1997 5M

Determine the transfer function of the system having the following state variable representation:

$$\dot{x} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -40 & -44 & -14\end{bmatrix}x + \begin{bmatrix}0\\ 1\\ 0\end{bmatrix}u$$

$$Y = \begin{bmatrix}0 & 1 & 0\end{bmatrix}x$$

Q9.95.1 MCQ 1995 1M

A system is described by the state equation $\dot{X} = AX + BU$. The output is given by $Y = CX$ where

$$A = \begin{bmatrix}-4 & -1\\ 3 & -1\end{bmatrix}, \quad B = \begin{bmatrix}1\\ 1\end{bmatrix}, \quad C = \begin{bmatrix}1 & 0\end{bmatrix}.$$

Transfer function $G(s)$ of the system is:

A. $\dfrac{s}{s^2+5s+7}$ **B.** $\dfrac{1}{s^2+5s+7}$

C. $\dfrac{s}{s^2+3s+2}$ **D.** $\dfrac{1}{s^2+3s+2}$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.95.2 and Q9.95.3 using the following data. These two questions together carry a total of **5 marks**. The state equation of a linear time-invariant system is given by

$$\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ -1 & -2\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}r(t)$$

Q9.95.2 NAT 1995

Find state transition $\phi(t)$ and the characteristic equation of the system.

Q9.95.3 NAT 1995

Determine the state vector $x_2(t)$ for $t = 0$ when $r(t) = U(t)$. Assume values of states initially to be zero.

Q9.94.1 MCQ 1994 1M

The matrix A of any state-space equations for the transfer function $C(s)/R(s)$ of the system, shown below in Figure, is

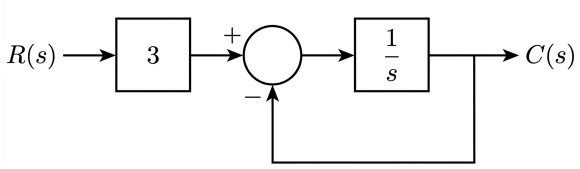


Fig. Q9.94.1

A. $\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$ **B.** $\begin{bmatrix}0 & 1\\ 0 & -1\end{bmatrix}$ **C.** $[-1]$ **D.** $[3]$

Q9.93.1 MCQ 1993

The transfer function for the state variable representation $\dot{x} = Ax + Bu$ and $y = Cx + Du$ is given by

A. $D + C[sI - A]^{-1}B$ **B.** $B[sI - A]^{-1}C + D$

C. $D[sI - A]^{-1}B + C$ **D.** $C[sI - A]^{-1}D + B$

Q9.93.2 MCQ 1993

Consider a second order system whose state space representation is of the form $\dot{x} = Ax + Bu$. If $x_1(t) = x_2(t)$, then system is

A. Controllable **B.** Uncontrollable

C. Observable **D.** Unstable

Q9.88.1 MCQ 1988

Given the following state-space description of a system

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-2 & 0\\ 0 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

The state-transition matrix will be

A. $\begin{bmatrix}e^{2t} & 0\\ 0 & e^{4t}\end{bmatrix}$ **B.** $\begin{bmatrix}e^{-2t} & 0\\ 0 & e^{4t}\end{bmatrix}$

C. $\begin{bmatrix}e^{-2t} & 0\\ 0 & e^{-4t}\end{bmatrix}$ **D.** $\begin{bmatrix}e^{2t} & 0\\ 0 & e^{-4t}\end{bmatrix}$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q9.26.1 D	Q9.26.2 C	Q9.25.3 3 to 3
Q9.23.1 2 to 2	Q9.21.1 1 to 1	Q9.19.1 A
Q9.18.1 C	Q9.17.1 D	Q9.17.2 1.280 to 1.287
Q9.16.1 5.9 to 6.1	Q9.15.1 B	Q9.15.2 A
Q9.14.1 C	Q9.14.2 A	Q9.14.3 C
Q9.14.4 A	Q9.13.1 A	Q9.13.2 A
Q9.12.1 D	Q9.10.1 C	Q9.09.1 C
Q9.09.2 B	Q9.08.1 D	Q9.08.2 A
Q9.06.1 B	Q9.05.1 A	Q9.05.2 C
Q9.04.1 A	Q9.03.1 A	Q9.03.2 A
Q9.02.1 C	Q9.02.2 B	Q9.02.3 B
Q9.02.4 *	Q9.01.1 A	Q9.00.1 *
Q9.00.2 *	Q9.00.3 *	Q9.98.1 *
Q9.97.1 *	Q9.95.1 A	Q9.95.2 *
Q9.95.3 *	Q9.94.1 C	Q9.93.1 A
Q9.93.2 B	Q9.88.1 C	

UNIT

X

Controllers and Compensators

Topics

1. Basics and Types of Controllers — P, PI, PD, PID
2. Lag Compensator
3. Lead Compensator
4. Lag-Lead Compensator
5. Design Problems on Controllers and Compensators

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Unit Snapshot*

Stream: EE	**Questions:** 18	**MCQ:** 15	**MSQ:** 0	**NAT:** 3
1M: 6	**2M:** 12	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2017)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q10.25.1 NAT 2025 2M

A controller $D(s)$ of the form $(1 + K_D s)$ is to be designed for the plant

$$G(s) = \frac{1000\sqrt{2}}{s(s+10)^2}$$

as shown in the figure.

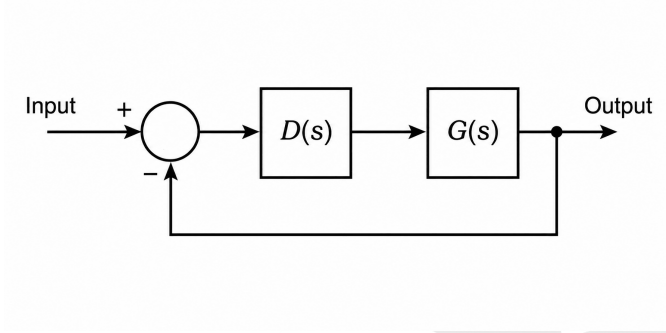


Fig. Q10.25.1

The value of K_D that yields a phase margin of 45° at the gain cross-over frequency of 10 rad/sec is

_____ (*round off to one decimal place*).

Q10.23.1 MCQ 2023 1M

Consider a unity-gain negative feedback system consisting of the plant $G(s)$ (given below) and a proportional-integral controller. Let the proportional gain and integral gain be 3 and 1, respectively. For a unit step reference input, the final values of the controller output and the plant output, respectively, are

$$G(s) = \frac{1}{s-1}$$

A. ∞, ∞ **B.** $1, 0$ **C.** $1, -1$ **D.** $-1, 1$

Q10.23.2 MCQ 2023 2M

Consider a lead compensator of the form

$$K(s) = \frac{1 + \frac{s}{a}}{1 + \frac{s}{\beta a}}, \quad \beta > 1,\ a > 0$$

The frequency at which this compensator produces maximum phase lead is 4 rad/s. At this frequency, the gain amplification provided by the controller, assuming asymptotic Bode-magnitude plot of $K(s)$, is 6 dB. The values of a, β, respectively, are

A. 1, 16 **B.** 2, 4 **C.** 3, 5 **D.** 2.66, 2.25

Q10.21.1 NAT 2021 2M

Consider a closed-loop system as shown.

$G_p(s) = \frac{14.4}{s(1+0.1s)}$ is the plant transfer function and $G_c(s) = 1$ is the compensator. For a unit-step input, the output response has damped oscillations. The damped natural frequency is

_____ rad/s. (*Round off to 2 decimal places*)

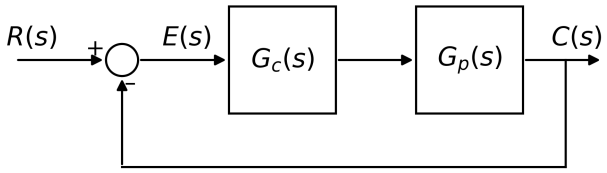


Fig. Q10.21.1

Q10.21.2 NAT 2021 2M

In the given figure, plant

$G_p(s) = \frac{2.2}{(1+0.1s)(1+0.4s)(1+1.2s)}$ and compensator $G_c(s) = K\left(\frac{1+T_1 s}{1+T_2 s}\right)$. The external disturbance input is $D(s)$. It is desired that when the disturbance is a unit step, the steady-state error should not exceed 0.1 unit. The minimum value of K is

_____. (*Round off to 2 decimal places*)

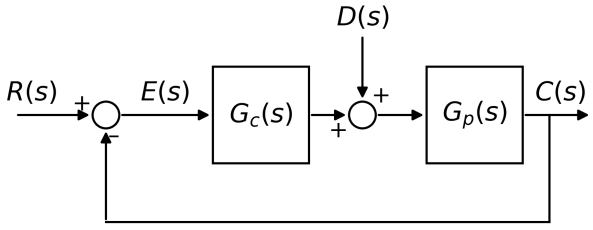


Fig. Q10.21.2

Q10.19.1 MCQ 2019 2M

The transfer function of a phase lead compensator is given by

$$D(s) = \frac{3\left(s + \frac{1}{3T}\right)}{\left(s + \frac{1}{T}\right)}.$$

The frequency (in rad/sec), at which $\angle D(j\omega)$ is maximum, is

A. $\sqrt{\frac{3}{T^2}}$ **B.** $\sqrt{\frac{1}{3T^2}}$ **C.** $\sqrt{3T}$ **D.** $\sqrt{3T^2}$

Q10.17.1 MCQ 2017 1M

The transfer function $C(s)$ of a compensator is given below.

$$C(s) = \frac{\left(1 + \frac{s}{0.1}\right)\left(1 + \frac{s}{100}\right)}{(1+s)\left(1 + \frac{s}{10}\right)}$$

The frequency range in which the phase (lead) introduced by the compensator reaches the maximum is

A. $0.1 < \omega < 1$ **B.** $1 < \omega < 10$
C. $10 < \omega < 100$ **D.** $\omega > 100$

Q10.14.1 MCQ 2014 2M

The magnitude Bode plot of a network is shown in the figure

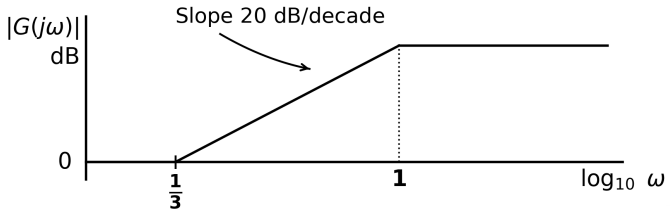


Fig. Q10.14.1

The maximum phase angle ϕ_m and the corresponding gain G_m respectively, are

A. $-30°$ and 1.73 dB **B.** $-30°$ and 4.77 dB
C. $+30°$ and 4.77 dB **D.** $+30°$ and 1.73 dB

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.12.1 and Q10.12.2 using the following data.
The transfer function of a compensator is given as

$$G_c(s) = \frac{s+a}{s+b}.$$

Q10.12.1 MCQ 2012 2M

$G_c(s)$ is a lead compensator if

A. $a = 1,\ b = 2$ **B.** $a = 3,\ b = 2$
C. $a = -3,\ b = -1$ **D.** $a = 3,\ b = 1$

Q10.12.2 MCQ 2012 2M

The phase of the above lead compensator is maximum at

A. $\sqrt{2}$ rad/s **B.** $\sqrt{3}$ rad/s
C. $\sqrt{6}$ rad/s **D.** $1/\sqrt{3}$ rad/s

Q10.08.1 MCQ 2008 2M

The transfer functions of two compensators are given below:

$$C_1 = \frac{10(s+1)}{(s+10)}, \qquad C_2 = \frac{s+10}{10(s+1)}$$

Which one of the following statements is correct?

A. C_1 is a lead compensator and C_2 is a lag compensator

B. C_1 is a lag compensator and C_2 is a lead compensator

C. Both C_1 and C_2 are lead compensators

D. Both C_1 and C_2 are lag compensators

Q10.07.1 MCQ 2007 2M

The system

$$\frac{900}{s(s+1)(s+9)}$$

is to be compensated such that its gain-crossover frequency becomes same as its uncompensated phase-crossover frequency and provides a 45° phase margin. To achieve this, one may use

A. a lag compensator that provides an attenuation of 20 dB and a phase lag of 45° at the frequency of $3\sqrt{3}$ rad/s

B. a lead compensator that provides an amplification of 20 dB and a phase lead of 45° at the frequency of 3 rad/s

C. a lag-lead compensator that provides an amplification of 20 dB and a phase lag of 45° at the frequency of $\sqrt{3}$ rad/s

D. a lag-lead compensator that provides an attenuation of 20 dB and a phase lead of 45° at the frequency of 3 rad/s

Q10.05.1 MCQ 2005 2M

If the compensated system shown in Figure below has a phase margin of 60° at the crossover frequency of 1 rad/sec, the value of the gain K is:

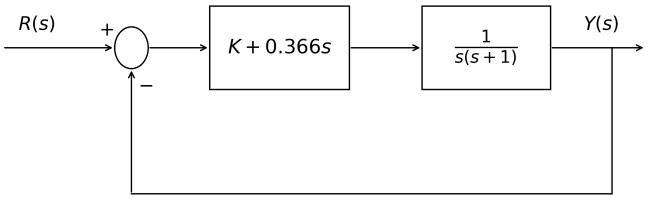


Fig. Q10.05.1

A. 0.366 **B.** 0.732 **C.** 1.366 **D.** 2.738

Q10.03.1 MCQ 2003 1M

A lead compensator used for a closed loop controller has the following transfer function

$$K\frac{\left(1+\frac{s}{a}\right)}{\left(1+\frac{s}{b}\right)}.$$

For such a lead compensator

A. $a < b$ **B.** $b < a$ **C.** $a > Kb$ **D.** $a < Kb$

Q10.00.1 MCQ 2000 2M

Maximum phase-lead of the compensator

$$D(s) = \frac{(0.5s+1)}{(0.05s+1)},$$

is

A. 52° at 4 rad/sec **B.** 52° at 10 rad/sec
C. 55° at 12 rad/sec **D.** None of the above

Q10.98.1 MCQ 1998 1M

The phase lead compensation is used to

A. increase rise time and decrease overshoot

B. decrease both rise time and overshoot

C. increase both rise time and overshoot

D. decrease rise time and increase overshoot

Q10.97.1 MCQ 1997 1M

Introduction of integral action in the forward path of a unity feedback system results in a

A. marginally stable system

B. system with no steady state error

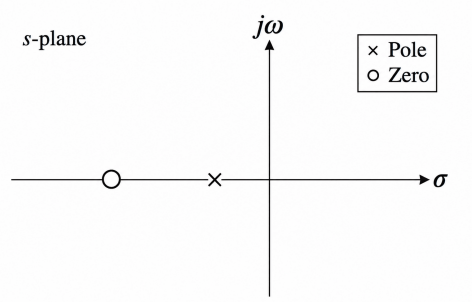
C. system with increased stability margin

D. system with better speed of response

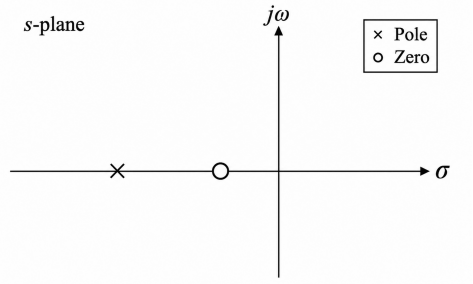
Q10.94.1 MCQ 1994 1M

The pole-zero configuration of a phase-lead compensator is given by

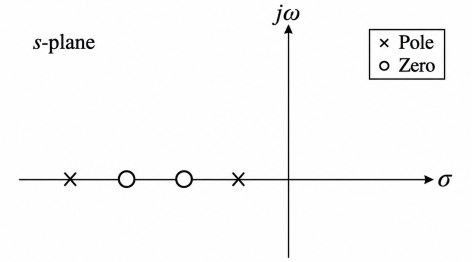
A.



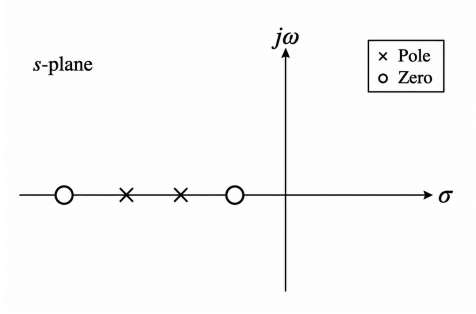
B.



C.



D.



Answer Key

Q10.25.1 0.1 to 0.1	Q10.23.1 D	Q10.23.2 B
Q10.21.1 10.80 to 11.00	Q10.21.2 9.50 to 9.60	Q10.19.1 B
Q10.17.1 A	Q10.14.1 C	Q10.12.1 A
Q10.12.2 A	Q10.08.1 A	Q10.07.1 D
Q10.05.1 C	Q10.03.1 A	Q10.00.1 D
Q10.98.1 B	Q10.97.1 B	Q10.94.1 B

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